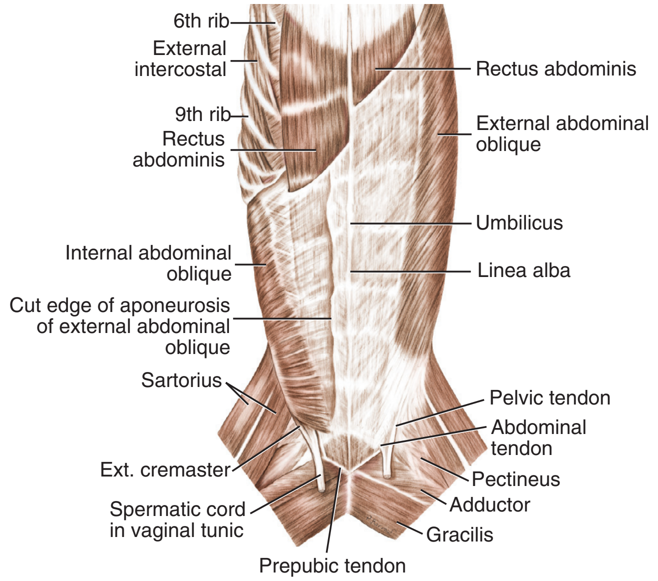

Figure 5-34 Superficial muscles of the ventral trunk. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

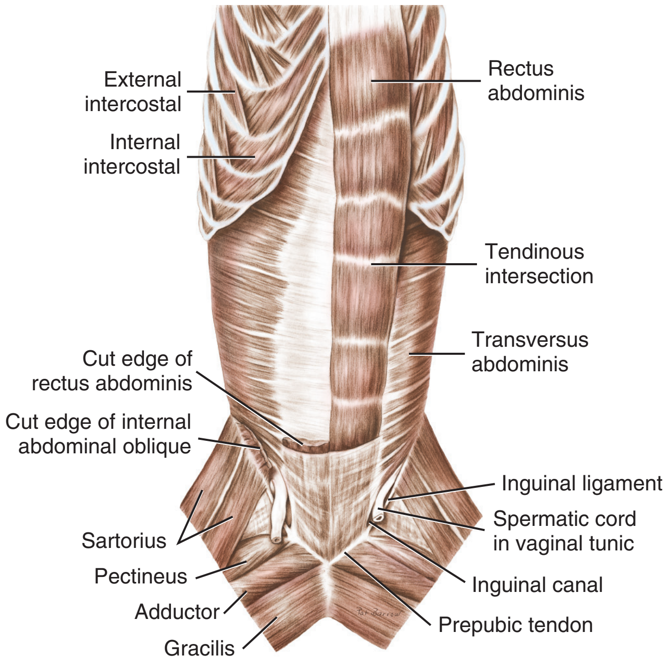

Figure 5-35 Deep muscles of the ventral trunk. (From Evans HE: *Miller's anatomy of the dog,* ed 4, St Louis, 2013, WB Saunders.)

Lymph Node Anatomy

Forelimb

Superficial cervical lymph nodes are located just cranial to the shoulder joint. They are deep to the cleidocephalicus and omotransversarius muscles and are on the lymph drainage path from the cutaneous area of the head, neck, and forelimb. The superficial cervical lymph nodes are almost always palpable. Axillary lymph nodes are located dorsal to the deep pectoral muscle and generally are not palpable unless they are abnormally enlarged. The deep structures of the forelimb, as well as lymph vessels from the thoracic wall, drain into these nodes.

Hindlimb

The popliteal lymph node lies in the fat just caudal to the stifle joint and at the caudal border of the biceps femoris muscle. This is the largest lymph node in the hindlimb. The hindlimb structures distal to the stifle joint drain into this lymph node. The femoral lymph node is not consistently present and is small when present. It lies in the fat deep to the deep medial femoral fascia at the distal part of the femoral triangle. The femoral vessels lie cranial to the femoral lymph node.

Trunk and Neck

The superficial inguinal lymph nodes are important lymph nodes in the trunk. In addition to the lymph nodes, trunk internal organ anatomy that is particularly relevant for rehabilitation includes the lung lobes, the anatomy of which is critical when administering postural drainage of individual lobes for dogs with respiratory insufficiency.[19]

Ligaments

Information about attachment, fiber direction, and motion restricted by the major forelimb, hindlimb, and spinal ligaments is noted in Web Appendix 2.[4,20] Listed order of the fiber direction in three planes is noted by the main orientation of the fibers first and follows in sequence for the second orientation of importance and then the third.

Forelimb

Figures 5-37 and 5-38 display ligament attachment site, location, and line of direction for the major ligaments of the forelimb. The canine shoulder joints have medial and lateral glenohumeral ligaments, which are mildly thickened regions of fibrous joint capsule. The transverse humeral retinaculum holds the tendon of the biceps brachii securely in place. The annular ligament holds the proximal radius securely to the ulna. The medial and lateral collateral ligaments of the elbow are substantial structures and contribute to elbow stability. Although there are no carpal collateral ligaments that span all the joints of the carpus, there are individual collateral ligaments at the various levels that are critical to carpal stability. Palmar ligament and fibrocartilaginous support of the carpus are vital for normal weight bearing.

Hindlimb

Figures 5-39 to 5-41 display hindlimb ligament attachment site, location, and line of direction for the major ligaments

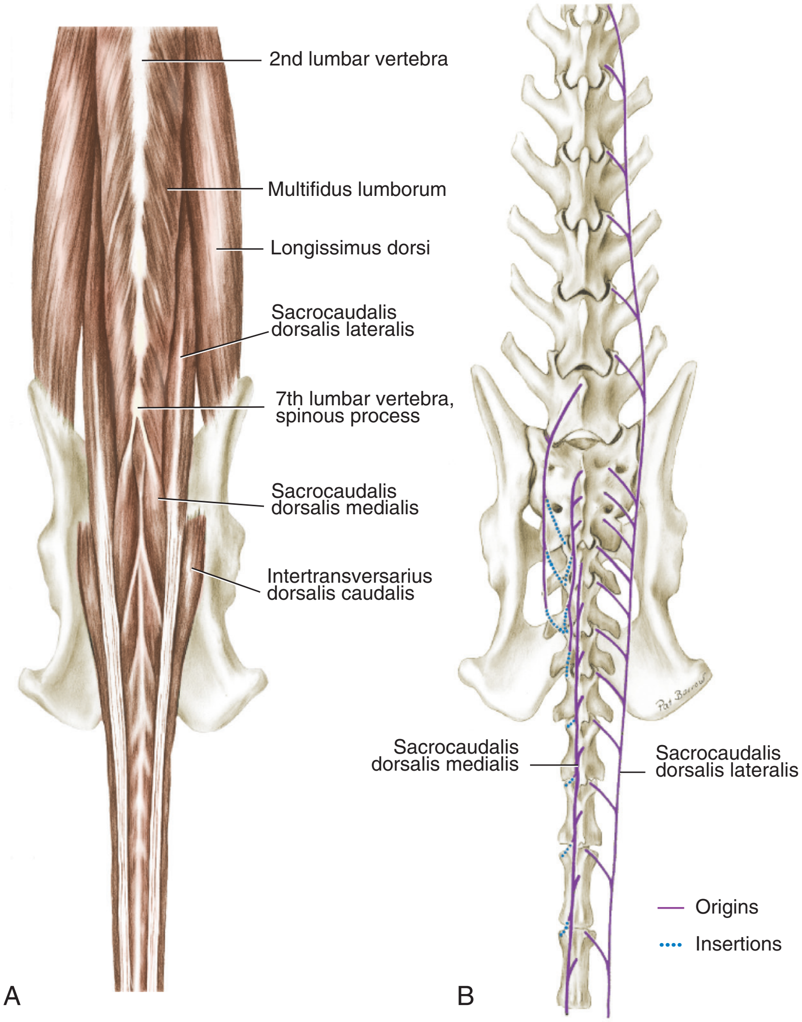

Figure 5-36 Muscles of the lumbocaudal region. **A,** Epaxial muscles, dorsal view. **B,** Diagram of skeleton and sacrocaudal muscle attachments, dorsal view. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

of the hindlimb. The iliofemoral, pubofemoral, and ischiofemoral ligaments are not present in the dog. The ligament of the femoral head does not contribute to hip stability as much as the capsule, but is important in the development of the hip. The transverse acetabular ligament traverses the acetabular notch and thereby completes the rim of the acetabulum. Without the transverse acetabular ligament, the hip may luxate ventrally. Although the sacrotuberous ligament is a pelvic ligament, rather than a hip-joint ligament, it is mentioned here because of its importance as one of the origins of the biceps femoris muscle.

The cranial (anterior) and caudal (posterior) cruciate ligaments are critical stabilizing ligaments that restrain stifle motion within physiologic limits. The cruciate ligaments are named by their cranial or caudal attachment to the tibia. Injury to the cranial cruciate ligament is common. The cranial cruciate ligament has different fiber bundles, the craniomedial and caudolateral bands. Throughout the normal range of motion, these bands accept varying stress loads depending on the stage of motion. With the stifle in moderate flexion, excessive stifle internal rotation stretches the cranial cruciate ligament under the medial femoral condyle, causing mechanical trauma to the ligament.

The medial and lateral collateral ligaments of the stifle joint are substantial and important to joint stability. The medial meniscus is attached to the tibia cranially and caudally, and to the medial collateral ligament. The lateral meniscus is attached to the tibia cranially and the femur caudally (through the meniscofemoral ligament) and is separated from the lateral collateral ligament. Although the menisci move with flexion and extension, the more fixed medial meniscus is less accommodating than the lateral if subluxation occurs with loss of the cranial cruciate ligament. Therefore the medial meniscus is often damaged in cases of cranial cruciate rupture.

The fibrous extensor retinaculum secures the patella. The contribution from the fascia lata to the extensor

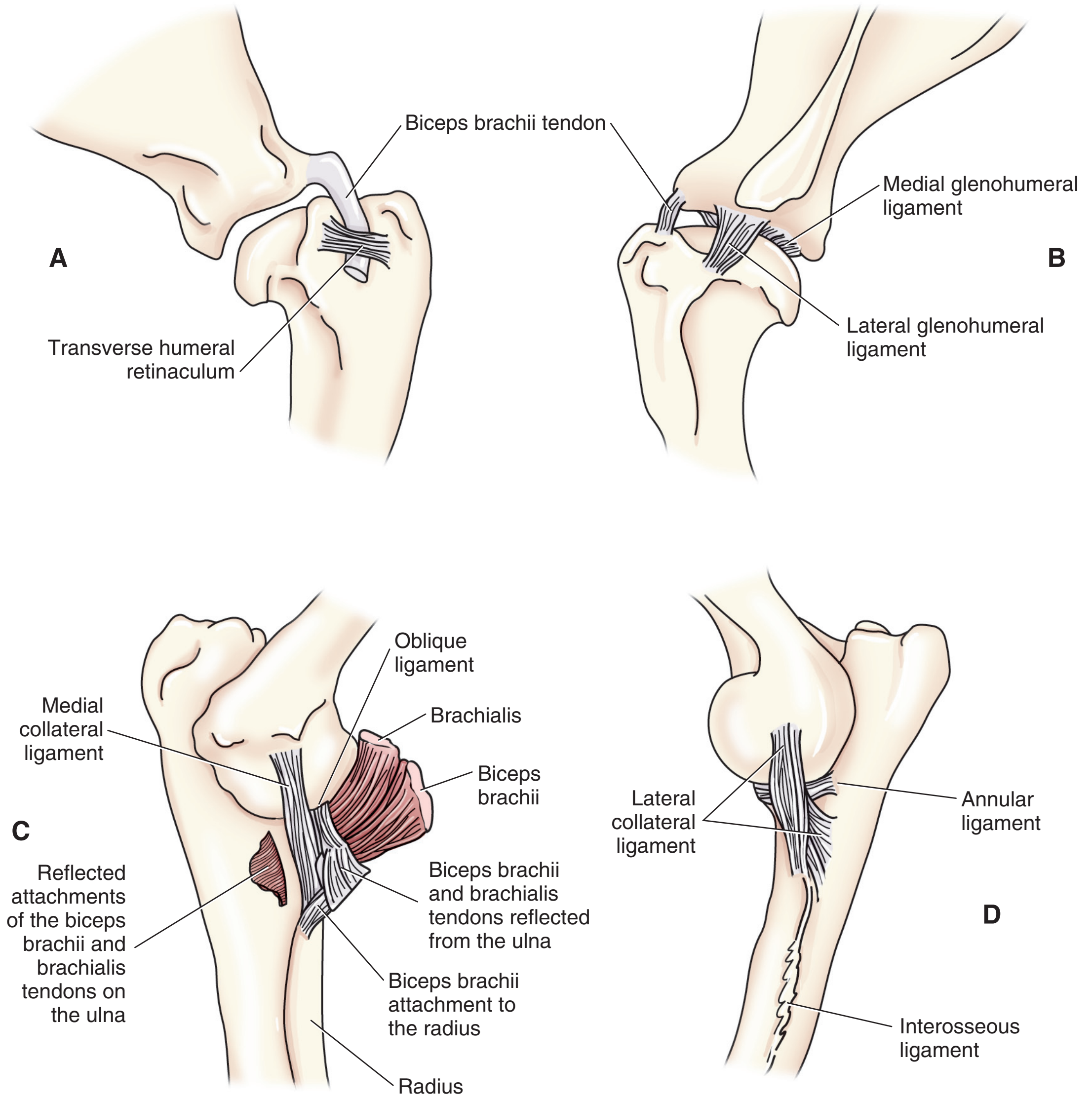

Figure 5-37 Ligaments of the medial (*A*) and lateral (*B*) left shoulder and medial (*C*) and lateral (*D*) left elbow in the dog. (Adapted from Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

retinaculum is particularly important in holding the patella in the trochlea. Femoropatellar ligaments are mildly thickened regions of the fibrous capsule and are observed only in large dogs. The tendon of the popliteus muscle separates the lateral collateral ligament from the lateral meniscus. The patellar ligament is a strong support between the patella and tibial tuberosity. There is no tendon from the quadriceps muscle to the patella. The quadriceps is attached to parapatellar fibrocartilage.

There are three joint or synovial pouches in the stifle, two femorotibial joint pouches, and one retropatellar joint pouch. The femorotibial joint pouches incorporate the articulations of the sesamoids of the gastrocnemius muscle, but exclude the cruciate ligaments. The lateral femorotibial pouch continues distally through the extensor groove as the tendon sheath for the tendon of the proximal long digital extensor muscle and also surrounds the tendon of the proximal attachment of the popliteus muscle. The tibiofibular joint has ligamentous support from the ligament of the fibular head, the interosseous membrane, and the cranial tibiofibular ligament.

Medial and lateral collateral ligaments of the tarsocrural joint provide stability to the tarsus. There are long and short portions of the collateral ligaments. Plantar ligaments are more developed and substantial in the dog than in the human because of the upright weight-bearing stance in humans. Ligaments in the hindpaw maintain what is described as a *rigid tarsocrural area* in dogs. The tendons coursing into the hindpaw are secured by the fascia pedis, which forms retinacula to stabilize tendons and superficial muscles of the hindpaw and secure them to all the projecting bones in the hindpaw.

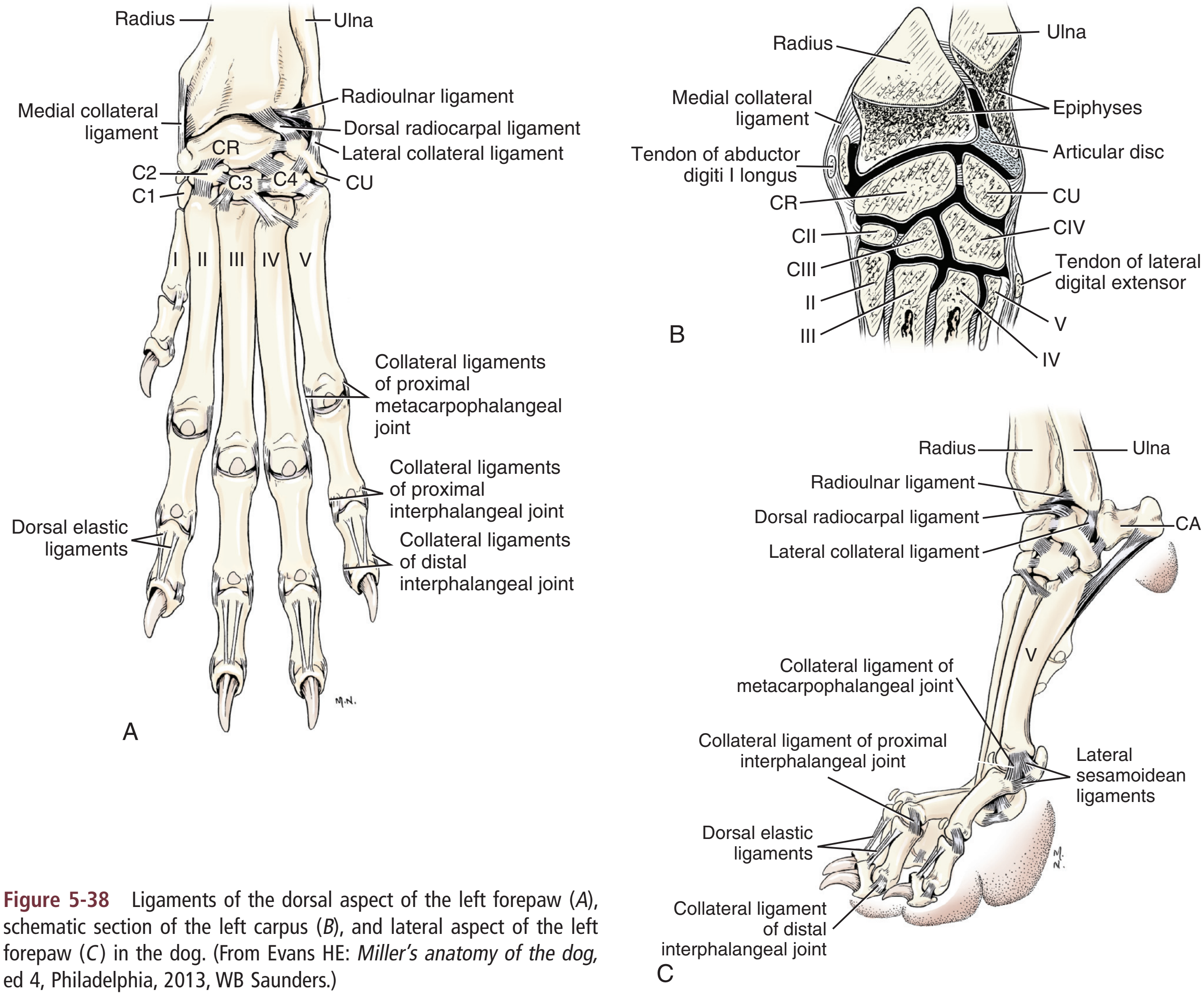

Figure 5-38 Ligaments of the dorsal aspect of the left forepaw (*A*), schematic section of the left carpus (*B*), and lateral aspect of the left forepaw (*C*) in the dog. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

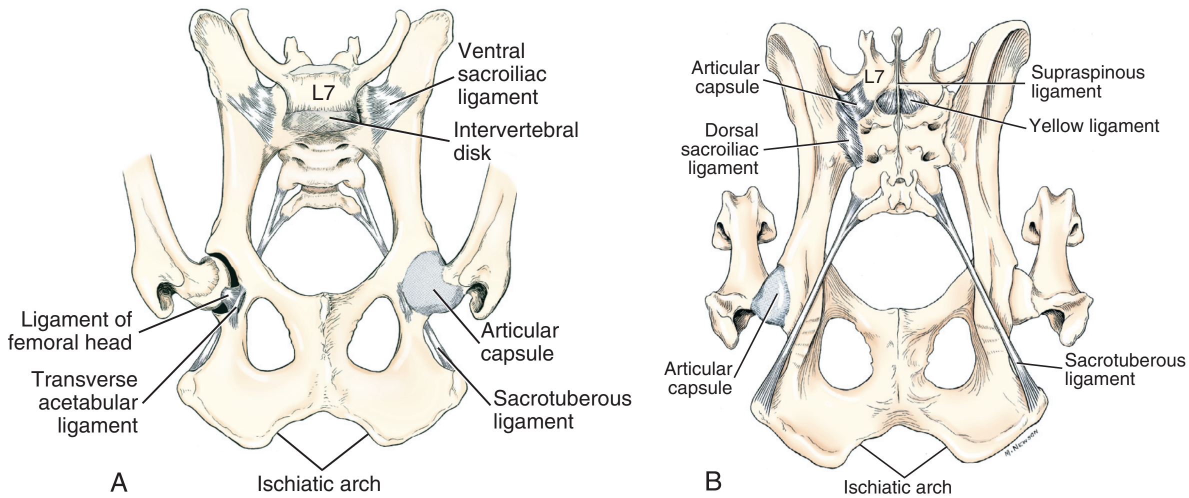

Figure 5-39 Ligaments of the hip and pelvis from a ventral (*A*) and dorsal (*B*) view. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

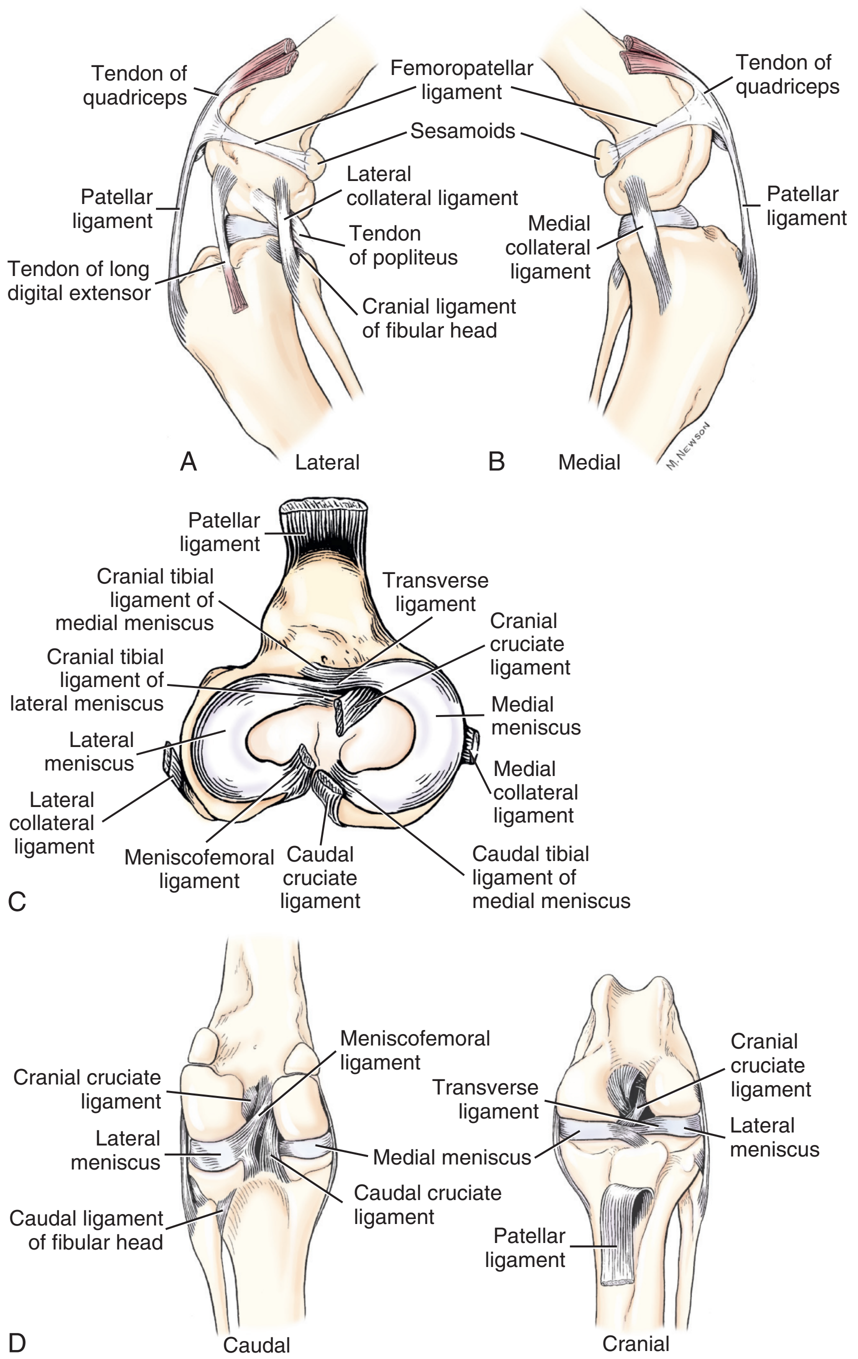

Figure 5-40 Ligaments of the left stifle in the dog from a lateral (*A*), medial (*B*), dorsal (*C*), caudal (*D*), and cranial (*E*) view. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

Spine

Figures 5-11, 5-12, 5-32, and 5-42 display ligament attachment sites, locations, and lines of direction for some ligaments of the spine. Spinal ligaments are extensive. Strong ligamentous support in the spine is thought to decrease the risk of disk herniation. The ligamentum nuchae, or nuchal ligament, is attached at the cranial and caudal extents of the cervical spine. This strong ligament acts as a tension band to oppose flexion of the head and neck and helps to hold the dog's head up with passive force, sparing muscle activity. This allows the dog relief from muscle fatigue.

General Nerve Supply

Forelimb

Figures 5-43 to 5-45 show the main nerves of the forelimb. The brachial plexus innervates the forelimb. It is formed by ventral rami C6 through T2. Sometimes C5 contributes to the brachial plexus. The nerves appear along the ventral border of the scalenus and enter the forelimb through the axillary space. The nerves of the canine brachial plexus include the suprascapular (C6-C7), subscapular (C6-C7), thoracodorsal (C6-C8), brachiocephalic (C6-C7), lateral

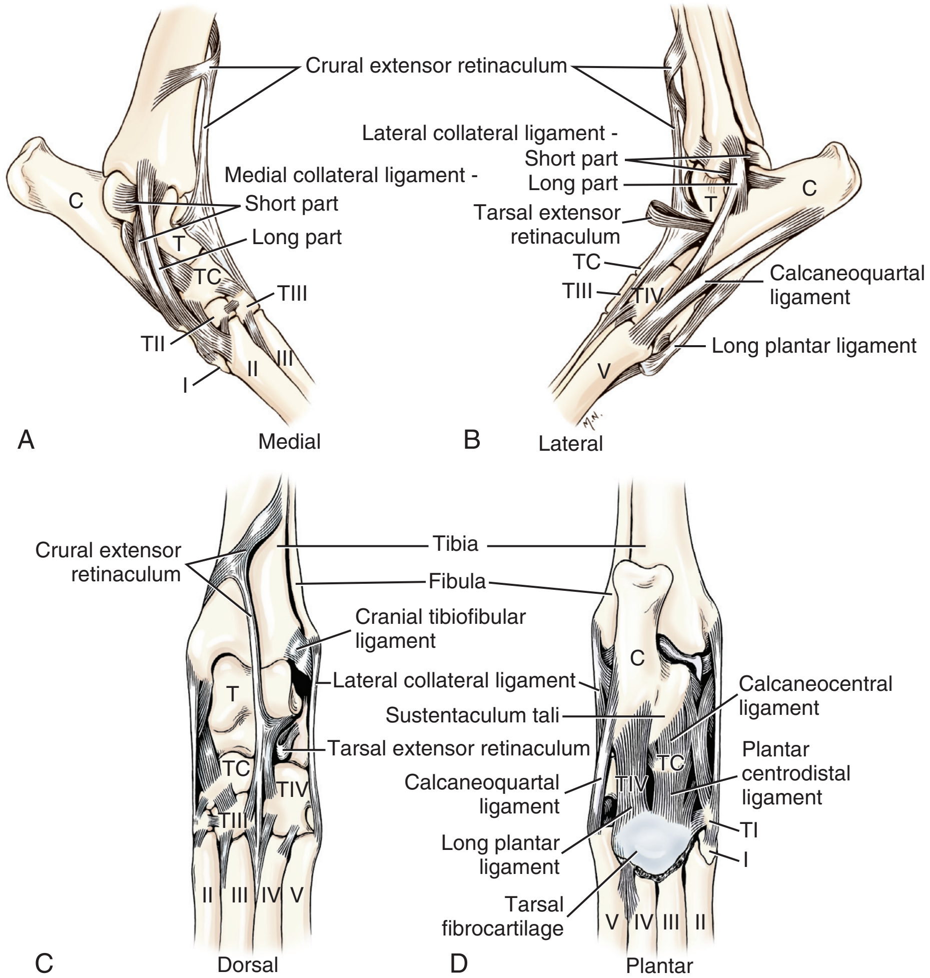

Figure 5-41 Ligaments of the left hindpaw in the dog from a medial (*A*), lateral (*B*), dorsal (*C*), and plantar (*D*) view. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

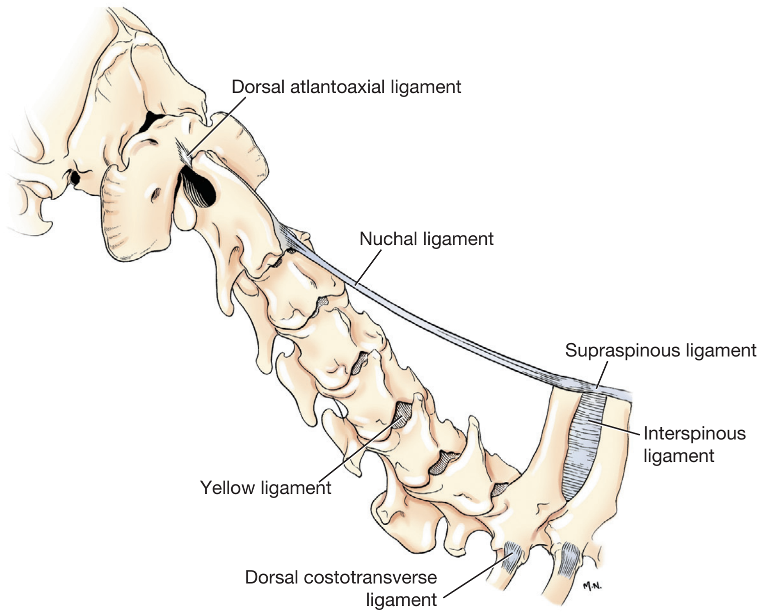

Figure 5-42 Nuchal ligament. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

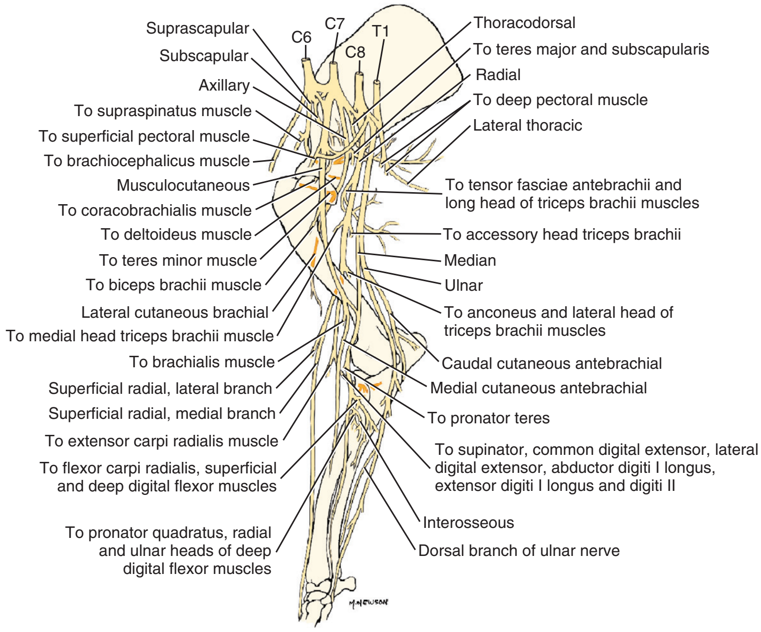

Figure 5-43 Main nerves of the right arm and forearm, medial aspect. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

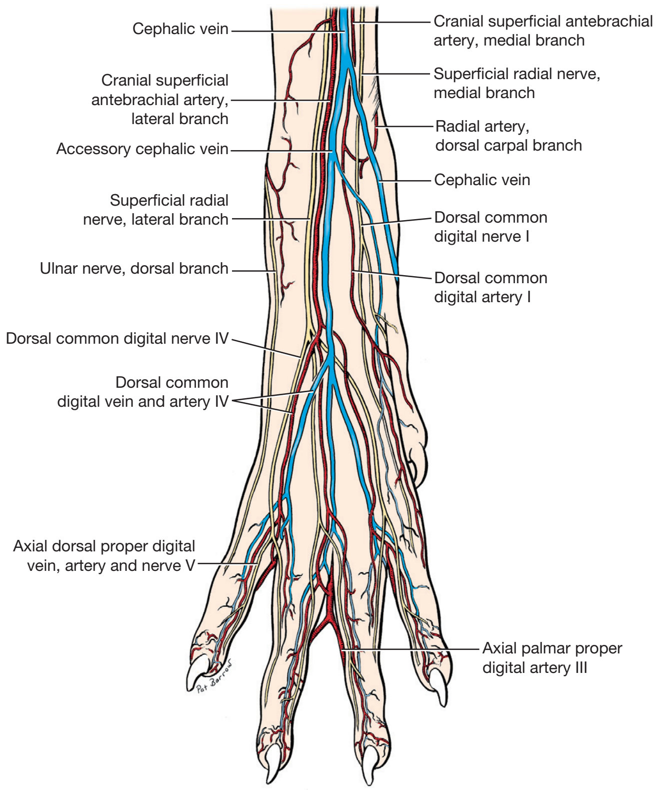

Figure 5-44 Nerves and arteries of the right forepaw, dorsal aspect. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

thoracic (C8-T1), caudal pectoral (C6-C8), musculocutaneous (C6-C8), axillary (C6-C8), radial (C6-T2), median (C7-T1), and ulnar (C8-T2) nerves. Innervations of muscles of the forelimb are included in Web Appendix 1.

Autonomous zones are areas of skin or cutaneous sensation that are innervated by only one peripheral nerve. Main front-limb autonomous zones of the brachial plexus are the regions of the axillary, musculocutaneous, radial, median, and ulnar nerves. Autonomous zones of the front limb and cranial trunk are displayed in Figure 5-46.

Hindlimb

Figures 5-47 to 5-49 show the main nerves of the hindlimb. The lumbosacral plexus innervates the hindlimb. This plexus can be divided into lumbar and sacral plexuses; however, there are always communications between the two plexuses. The plexus consists of ventral rami from the caudal five lumbar, or L, nerves and the three sacral, or S, nerves. There are seven lumbar nerves and three sacral nerves. The lumbosacral trunk is the largest and most important part of the lumbosacral plexus and becomes the sciatic nerve outside the pelvis.

The lumbar plexus is generally formed by ventral rami from L3-L5, sometimes L2-L6. The nerves of the lumbar plexus include the ilioinguinal from L3, genitofemoral from L3 and L4, lateral cutaneous femoral from L3 and L4, femoral from L4-L6, saphenous from L4-L6, and obturator from L4-L6 nerves. Innervations of the rear limb muscles are included in Web Appendix 1.

The sacral plexus is formed by ventral rami from L6-S3. The nerves of the sacral plexus that relate to the hindlimb include the cranial gluteal from L6-S1; caudal gluteal from L7 (S1); caudal cutaneous femoral from S1-S2[21]; sciatic from

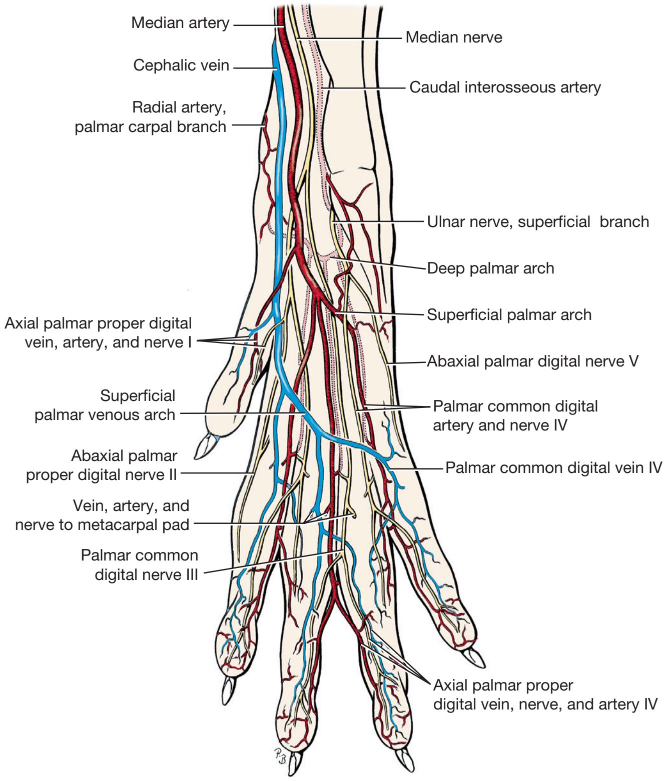

Figure 5-45 Nerves and arteries of the right forepaw, palmar aspect. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

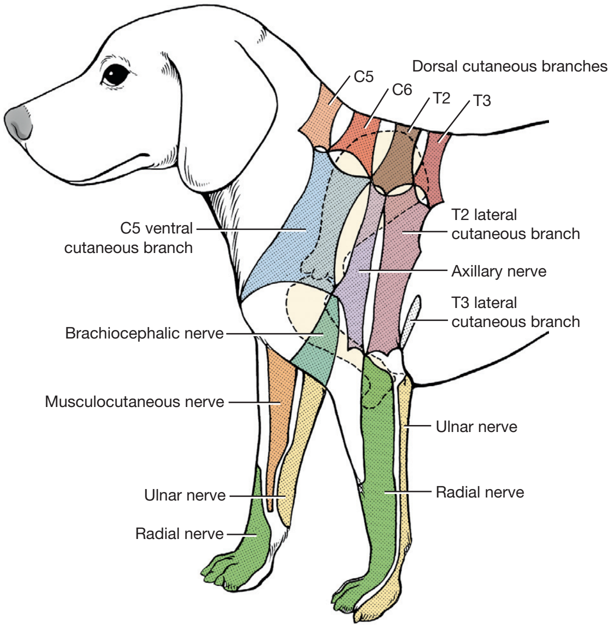

Figure 5-46 Autonomous zones of cutaneous innervation of the cranial trunk and forelimb. (After Kitchell RL et al: Electrophysiologic studies of cutaneous nerves of the thoracic limbs of the dog. *Am J Vet Res* 41:61-76,1980. In Evans HE, deLahunta A: *Miller's guide to the dissection of the dog,* ed. 7, Philadelphia, 2010, WB Saunders.)

L6-S1; common, superficial, and deep peroneal from L6-L7; tibial from L6-S1; proximal caudal cutaneous sural from L6-S1; and lateral cutaneous sural from L6-L7. In the hindpaw the dorsal metatarsal nerves are continuations of the deep peroneal nerve, and the medial plantar and lateral plantar nerves are the distal continuations of the tibial nerve. Innervations of muscles of the hindlimb are included in Web Appendix 1.

Main hindlimb autonomous zones are the regions of distribution of the genitofemoral, femoral, lateral femoral cutaneous, saphenous, caudal cutaneous femoral, sciatic, tibial, and peroneal nerves. Autonomous zones of the hindlimb and caudal trunk are displayed in Figure 5-50.

Trunk and Neck

The superficial muscles of the dorsal and lateral neck and trunk are innervated by the brachial plexus for muscles near the forelimb and by the lumbosacral plexus for muscles near the hindlimb. Most of the deep musculature of the neck and trunk is innervated segmentally by dorsal rami of spinal nerves (see Web Appendix 1). These deep muscles on the dorsal neck and trunk are epaxial muscles; that is, they developed in the dorsal neck or trunk area and received muscular innervation locally from the dorsal rami. The advantage of the segmental innervation from many nerves is that there is less chance of a strength deficit from damage to any one nerve.

The hypaxial muscles of the lateral and ventral abdominal wall developed in the lateral and ventral abdomen and received innervation from the ventral rami (i.e., the branch

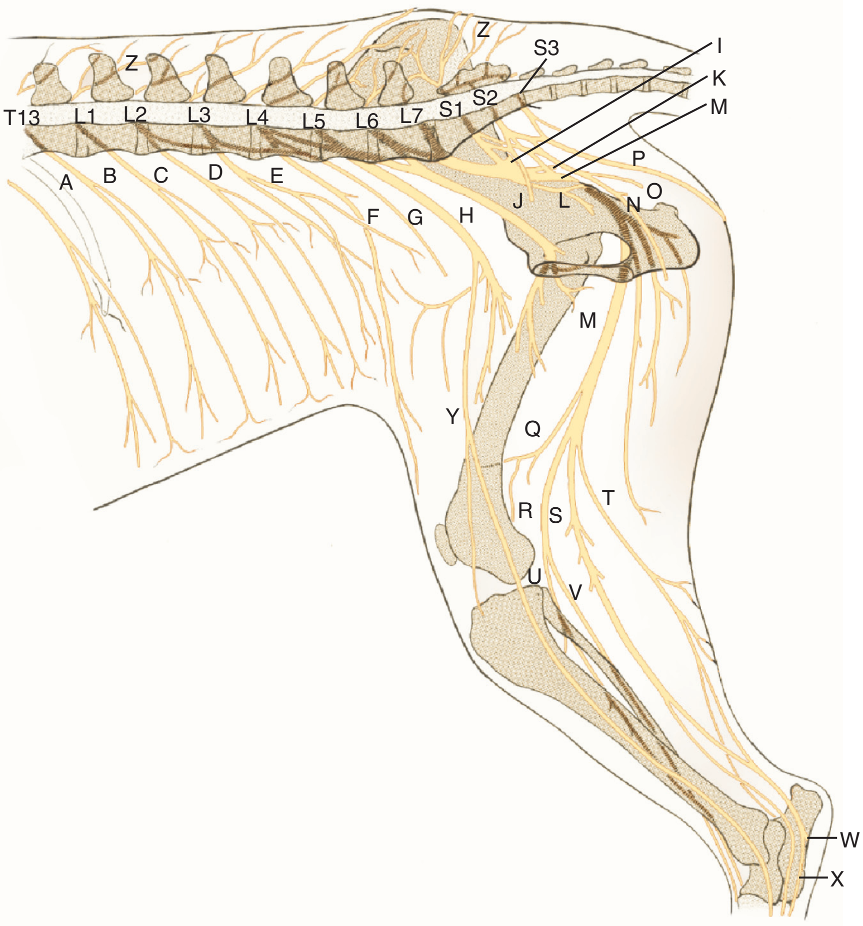

Figure 5-47 Schematic medial view of lumbar and sacral nerves. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

A. Thirteenth thoracic nerve, ventral branch
B. Cranial iliohypogastric nerve
C. Caudal iliohypogastric nerve
D. Ilioinguinal nerve
E. Lateral cutaneous femoral nerve
F. Genitofemoral nerve
G. Femoral nerve
H. Obturator nerve
I. Cranial gluteal nerve
J. Pelvic nerve
K. Caudal gluteal nerve
L. Nerve to mm. obturator internus, gemelli, and quadratus femoris
M. Ischiatic nerve
N. Pudendal nerve
O. Perineal nerve
P. Caudal cutaneous femoral nerve
Q. Lateral cutaneous sural nerve
R. Common fibular nerve
S. Tibial nerve
T. Caudal cutaneous sural nerve
U. Deep fibular nerve
V. Superficial fibular nerve
W. Lateral plantar nerve
X. Medial plantar nerve
Y. Saphenous nerve
Z. Dorsal branches of lumbar and sacral nerves

of the spinal nerve that traveled laterally and ventrally and innervated the local structures).

The spinal nerves exit between the vertebrae through the intervertebral foramina. All of the spinal nerves, except the cervical nerves, exit the intervertebral foramen caudal to the vertebra of the same number. In the cervical region, there are eight cervical nerves and seven cervical vertebrae. C2-C7 exit the intervertebral foramen cranial to the vertebra of the same number. C1 emerges through the lateral vertebral foramen of the arch of the atlas. C8 exits the intervertebral foramen caudal to C7.

Neck and trunk autonomous zones are circumferential strips of skin sensation around the dorsal neck and trunk. Because of extensive overlap in cutaneous sensation, a sensory deficit does not often occur unless a minimum of two consecutive spinal nerves are damaged.

General Blood Supply

Forelimb

Primary blood vessels to the dog's forelimb are the subclavian, axillary, brachial, common interosseous, and median arteries (Figure 5-51). The axillary artery and branches and the external thoracic, lateral thoracic, subscapular, and cranial and caudal circumflex humeral supply the scapular and shoulder area. The brachial artery and branches, deep brachial, bicipital, collateral ulnar, superficial brachial,

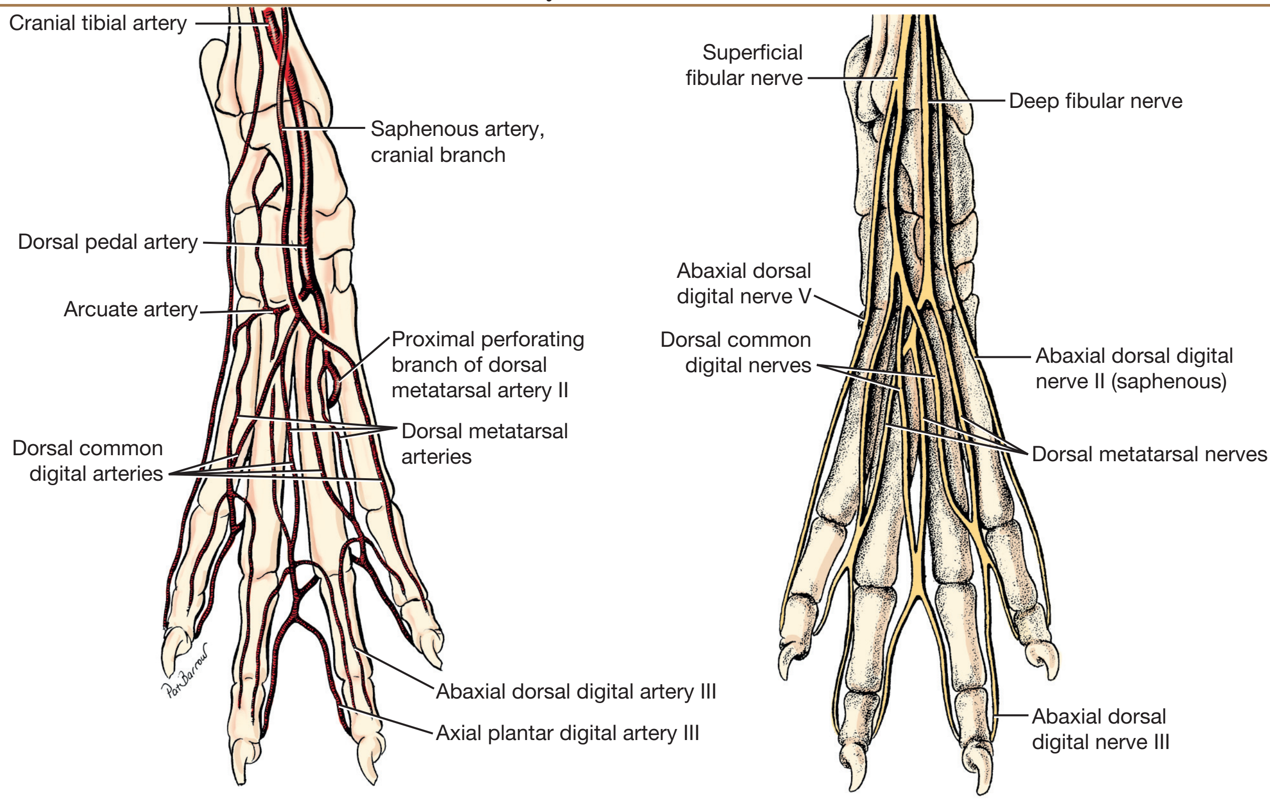

Figure 5-48 Nerves and arteries of the right hindpaw, dorsal aspect. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

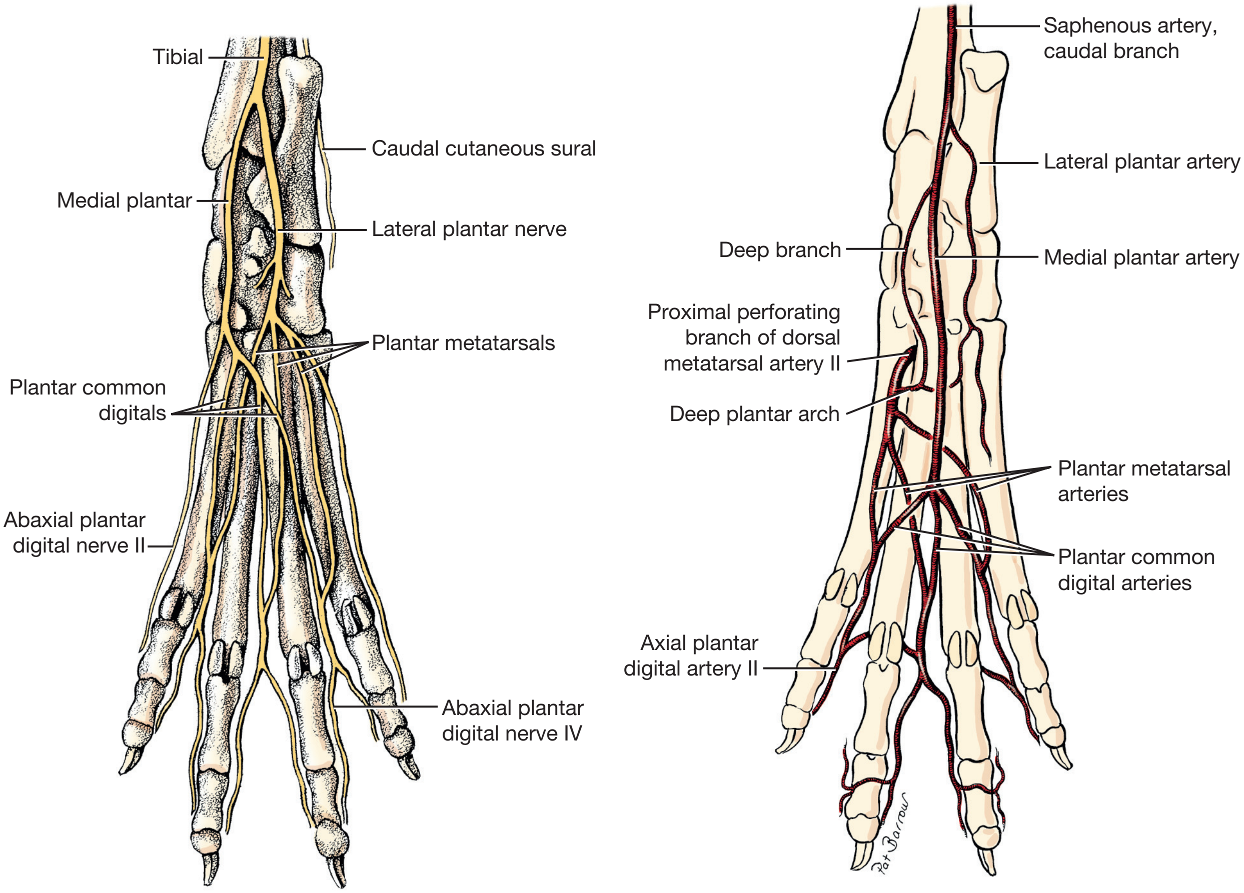

Figure 5-49 Nerves and arteries of the right hindpaw, plantar aspect. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

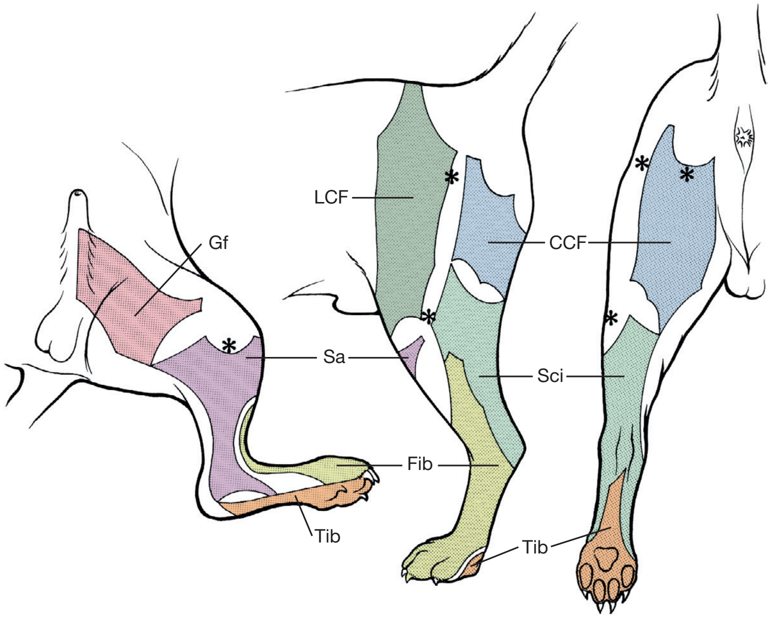

Figure 5-50 Autonomous zones of cutaneous innervation of the caudal trunk and hindlimb. *Asterisks* indicate palpable bony landmarks (medial and lateral tibial condyles, greater trochanter, and lateral end of the ischiatic or ischial tuberosity). The sciatic nerve autonomous zone is for lesions proximal to the greater trochanter and includes the zones for the peroneal and tibial nerves. For sciatic nerve lesions caudal to the femur, the autonomous zone varies, depending on how many of its cutaneous branches are affected. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

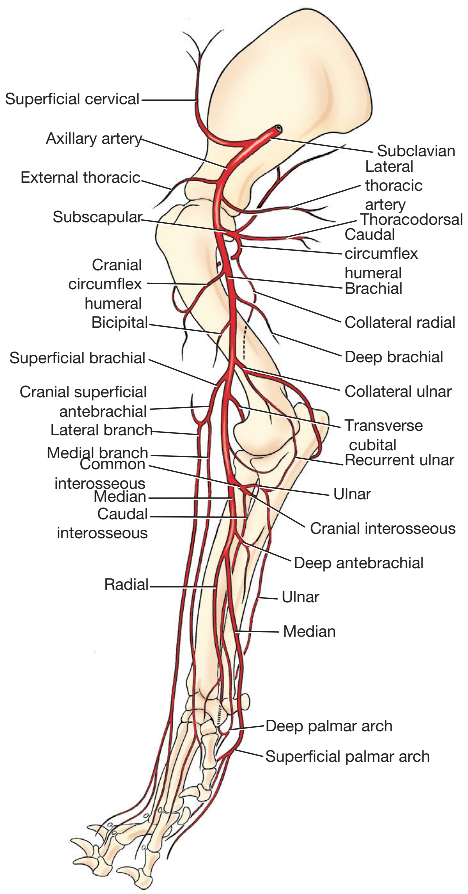

Figure 5-51 Arteries of the right forelimb, medial view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

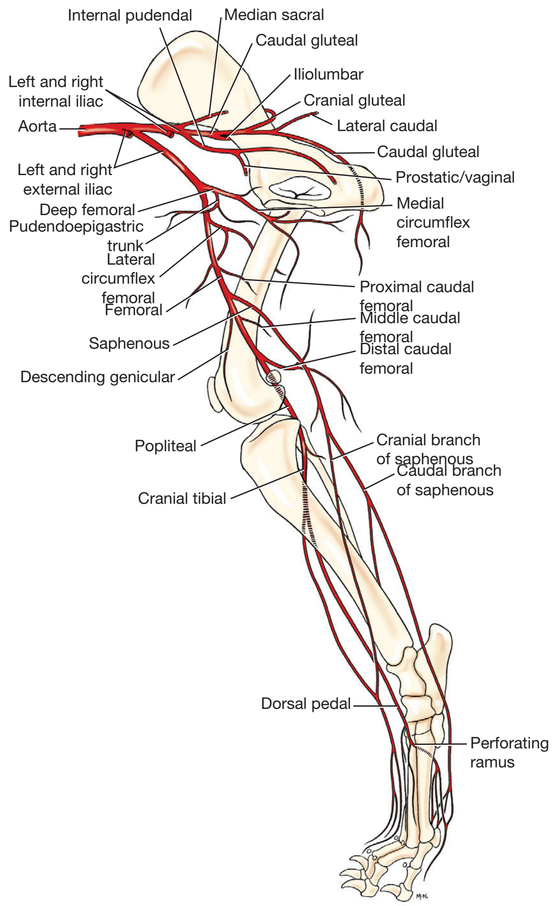

Figure 5-52 Arteries of the right hindlimb, medial view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

and common interosseous supply the arm and forearm. Common interosseous branches are the ulnar, cranial, and caudal interosseous arteries, which supply the deep forearm. The median artery, a continuation of the brachial artery, has two branches, the superficial palmar arch, which is the distal continuation of the median artery, and the radial artery. Arterial pulses are not easily palpable in the front limb and are not commonly used to assess cardiovascular function.

Hindlimb

The abdominal aorta divides into the common iliac arteries. The two branches of the common iliac arteries are the internal iliac artery, supplying the pelvic area, and the external iliac, which supplies the lower limb (Figure 5-52). The internal iliac artery supplies the caudal thigh via the caudal gluteal artery. The external iliac branches to the deep femoral, which gives rise to the medial circumflex femoral and the femoral arteries. The femoral artery gives rise to the lateral circumflex femoral, saphenous, descending genicular, distal caudal femoral, and popliteal arteries. Arterial pulses are palpable in the femoral artery in the inguinal region. From the caudal stifle area, the femoral artery becomes the popliteal, which divides into the cranial tibial, which continues as the dorsal pedal and arcuate arteries supplying the dorsal hindpaw, and the caudal tibial or posterior tibial arteries. The dorsal pedal artery is the principal blood supply to the hindpaw. The saphenous artery from the femoral sends genicular branches to the stifle before dividing into a superficial and caudal branch. The superficial branch anastomoses with the superficial branch of the cranial tibial artery to form a series of dorsal metatarsal, common digital, and axial and abaxial proper dorsal digital arteries coursing between the toes. The caudal branch of the saphenous artery forms the larger medial and a smaller lateral plantar arteries, and from these are formed a series of plantar common digital arteries. A perforating branch of dorsal metatarsal artery II, which is a branch of the arcuate artery, forms the deep plantar arch, from which form the plantar metatarsal arteries. The plantar metatarsal arteries join the plantar common digital arteries to form axial plantar digital arteries, which travel on the axial sides of the toes.

The cranial branch of the lateral saphenous vein is commonly a site for venipuncture. It is located on the dorsal tarsus and courses proximocaudally along the lateral surface of the leg.

Palpation of the Dog

Structures that are sufficiently superficial may be palpated. Palpation of structures, linked with knowledge of location of structures and the variation of feel with pathologic findings, is an important skill in accurate palpation. Palpation is more difficult in some dogs, depending on the type and thickness of hair, skin, and subcutaneous fat. Structures that can be palpated in dogs are numerous. Almost all figures in this chapter illustrate some palpable structures.

Forelimb

Bony Landmarks

The bony landmarks that are easily palpated include the following:

- *Scapula:* Borders, spine, acromion
- *Humerus:* Greater tubercle, intertubercular groove, deltoid tuberosity, medial and lateral epicondyles
- *Radius:* Head, medial styloid process
- *Ulna:* Olecranon, lateral styloid process

The medial coronoid process is usually covered by too much soft tissue to be palpable, but the location can be painful on palpation if the coronoid is fragmented.

- *Carpus and digits:* Accessory carpal bone, metacarpals, phalanges, metacarpophalangeal and interphalangeal joints, metacarpal pad in relation to the metacarpophalangeal joints, digital pads in relation to the distal interphalangeal joints

Joints

Joint motion can be palpated and observed with flexion and extension of the shoulder, elbow, carpus, and the metacarpophalangeal and interphalangeal joints.

Muscles and Ligaments

The forelimb muscles to be palpated include the following:

- *Scapulothoracic:* Trapezius, rhomboideus
- *Glenohumeral:* Superficial pectoral, deep pectoral, brachiocephalicus, latissimus dorsi, supraspinatus, infraspinatus, deltoideus, long head of triceps, biceps brachii tendon in the intertubercular groove, biceps brachii medially and brachialis laterally on the distal cranial humerus.
- *Elbow:* Triceps brachii and tendon of insertion
- *Forepaw:* Flexor muscles on the medial humeral epicondyle, distal attachments of the flexor carpi ulnaris and ulnaris lateralis on the accessory carpal, flexor carpi radialis as it tightens with carpal extension, extensor muscles, tendons of the extensor carpi radialis and the common digital extensor
- *Digit:* Tendons of the superficial digital flexors at the carpal canal, and digital flexor tendons and extensor tendons as groups
- *Ligaments:* Although the medial and lateral collateral ligaments of the elbow, annular ligament, short radial collateral, and short ulnar collateral ligaments of the carpal joint are not palpable, their integrity can be evaluated by applying controlled stresses to the joints.

Nerves

The ulnar nerve may be palpable caudal to the medial epicondyle. The median nerve is located just cranial to the humeral medial epicondyle where it is accessible for nerve stimulation but is not distinctly palpable.

Arteries

The brachial artery is located just cranial to the humeral medial epicondyle but may not be palpable. The median artery passes deep to the flexor carpi radialis at the carpal joint, where the pulse may be palpable.

Veins

The cephalic vein becomes prominent for palpation with compression of the cranial aspect of the proximal antebrachium.

Lymph Nodes

The superficial cervical lymph nodes can be palpated cranial to the shoulder.

Hindlimb

Bony Landmarks

The bony landmarks to be palpated include the following:

- *Pelvis:* Iliac crests, sacrum, cranial dorsal and ventral iliac spines, ischiatic tuberosities, ischiatic arch
- *Femur:* Greater trochanter, trochlea of the femur and patella that articulates with the trochlea, femoral condyles
- *Tibia:* Tibial condyles, tibial tuberosity, cranial border, medial side of the body of the tibia, medial malleolus
- *Fibula:* Head of the fibula, lateral malleolus
- *Tarsus:* Trochlea of the talus and lateral ridge, calcaneus, phalanges, metatarsal pad in relation to the metatarsophalangeal joints, digital pad in relation to the distal interphalangeal joints

Cartilage

The parapatellar fibrocartilages sometimes can be palpated.

Joints

Joint motion can be palpated and observed in the hip, stifle, tarsocrural, metatarsophalangeal, and interphalangeal joints.

Muscles and Ligaments

The hindlimb muscles to be palpated include the following:

- *Hip:* Superficial gluteal, middle gluteal, sartorius, biceps femoris, semitendinosus, semimembranosus, gracilis, and pectineus

In the caudal thigh, the biceps femoris, semitendinosus, semimembranosus, and gracilis are palpated in sequence from lateral to medial.

- *Stifle:* Vastus medialis, rectus femoris, vastus lateralis, biceps femoris, semitendinosus, semimembranosus, medial and lateral gastrocnemius and their associated sesamoid bones (fabellae), gracilis
- *Tarsus:* Craniolateral muscle group that forms the tarsocrural flexors and digital extensors, gastrocnemius muscles, common calcaneal tendon, superficial digital flexor tendon, and portions of the biceps femoris, semitendinosus, and gracilis tendons
- *Digits:* Digital flexors and extensors as groups, digital flexors in the caudal leg and plantar tarsus
- *Ligaments:* The medial collateral and lateral collateral ligaments of the stifle and tarsocrural joints, and the patellar ligament, are palpable. The extensor retinaculum may be palpable.

Nerves

The common peroneal nerve is palpable on the lateral surface of the proximal fibula. The tibial nerve is palpable cranial to the common calcaneal tendon and proximal to the tarsus.

Arteries

The femoral artery is palpated in the area of the medial femoral triangle and just cranial to the pectineus muscle. A pulse is palpable where the cranial branch of the saphenous artery typically crosses the medial side of the middle third of the tibia. The dorsal pedal artery can be palpated on the dorsal tarsus, often between the proximal ends of the second and third metatarsal bones.

Veins

The lateral saphenous vein is palpable and is a common site for venipuncture.

Lymph Nodes

The popliteal lymph node, between the distal ends of the biceps femoris and the semitendinosus, is palpable.

Trunk

Bony Areas

Palpable bony areas of the trunk include spinous processes of thoracic and lumbar vertebrae, transverse processes of the lumbar vertebrae, wings or lateral masses of atlas, spine of the axis, transverse processes of C3-C6, T13 spinous process from location of rib 13, cranial dorsal iliac spine, rib expansion with breathing, iliac crests and L7 spinous process between the iliac crests, sternum from manubrium to the xiphoid process, ribs.

Joints

Joint motion may be palpated and observed during intervertebral joint motion.

Muscles

The cervical and trunk extensor muscle mass and the rectus abdominis muscle are palpable.

Soft Tissue

The lumbosacral space caudal to the last lumbar spinous process is palpable.

Nerves

No nerves are readily palpable in the neck and trunk.

Arteries

The heartbeat can be felt over the left ventral thorax, and the pulse can be felt from the carotid artery. No veins are palpable in the neck and trunk, but with compression at the thoracic inlet, the external jugular vein is palpable.

Lymph Nodes

The inguinal lymph nodes are palpable.

Summary

This chapter presents canine anatomy in the context of association learning and the relevant anatomy for movement rehabilitation. Some of the relevant biomechanics that enhance association learning have been included. Knowledge of relevant anatomy and biomechanics is important in understanding movement function and dysfunction in animals.

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6 Tissue Healing: Tendons, Ligaments, Bone, Muscles, and Cartilage

Andrea Henderson and Darryl Millis

Injury to tissue initiates a complex series of events involving many cellular and biochemical responses that ultimately result in wound healing. The series of events depends on the severity of the injury and the tissues involved. The goal is regeneration or repair of the injured or traumatized tissue. Bone generally restores tissue architecture by regeneration of the normal cell population while other connective tissues respond by repair of the injured area.[1] The wound repair response is generally described by three major phases: inflammation, reparative (proliferative or fibroblastic), and remodeling (maturation).[1-3] These phases of healing overlap, and their sequence and timing vary with the tissue involved and the severity of the injury. The focus of this chapter is the basic wound healing response as well as the specific healing patterns of bone, muscle, tendon, ligament, and cartilage.

General Wound Healing

Inflammatory Phase

Most tissues undergo an initial inflammatory phase that involves an acute vascular response followed by cellular infiltration.[1,3] The immediate vascular component of the response is centered around hemostasis in the wound. Blood vessel disruption within the wound allows extravascular movement of blood elements and exposure of subendothelial collagen to platelets, causing activation of the coagulation cascade.[1,3-5] The result is formation of a fibrin network to support the hemostatic plug and act as a scaffold for cellular infiltration.

The cellular aspect of the inflammatory phase follows the initial vascular response. The release of cytokines associated with the hemostatic plug and increased vascular permeability incites the chemotaxis of cells into the wound environment (Figure 6-1). Neutrophils are the first cells migrating into the area, appearing by 6 hours after injury and peaking over a 2- to 3-day period.[1,3,5] The roles of the neutrophil are initial debridement and phagocytosis of microorganisms, thereby minimizing the potential for infection. The extent of neutrophil action within the wound depends on the severity of the wound and degree of contamination. The contribution by neutrophils in a noncontaminated wound is not essential for normal wound healing.[1,5]

Macrophages appear within the wound approximately 24 to 48 hours after neutrophil migration.[1] The influx of macrophages into the wound appears key to the transition from the inflammatory phase to the reparative phase and appropriate wound repair.[1,3] They are central to five major functions: phagocytosis, wound debridement, matrix synthesis regulation, cell recruitment and activation, and angiogenesis.[3] Phagocytosis and wound debridement occur with the release of oxygen radicals, nitric oxide, and collagenases.[3] Working in conjunction with neutrophils, macrophages create an optimal environment for entering the reparative (fibroblastic) phase of wound healing. This activity is at a maximum in the first 3 to 4 days after injury. In addition, macrophages release a host of cytokines, growth factors, prostaglandins, and enzymes that subsequently activate and mediate angiogenesis and fibroplasia.[3] The macrophage appears to be the central figure in control of cellular and biochemical events in general wound healing.

The presence of macrophages also attracts and activates lymphocytes, whose function remains controversial.[1,3] Lymphocytes secrete lymphokines such as interferons and interleukins that stimulate fibroblast migration and collagen synthesis.[1,3,6] Their activity appears to peak approximately 6 days after injury.[1,5]

Reparative Phase

The reparative (proliferative or fibroblastic) phase of wound healing is characterized by the cellular response of endothelial cells and fibroblasts.[1,3] Fibroblast proliferation and migration predominate and are followed by matrix synthesis, including collagen, elastin, and proteoglycans. The fibroblasts begin appearing in the wound within 3 days of injury; however, there is a lag phase of 2 to 3 days before initiation of collagen production with maximal proteoglycan synthesis not occurring until 2 weeks after injury.[5] Matrix synthesis increases over the following few weeks with concurrent increases in tensile strength.[1]

In addition, endothelial cells adjacent to the wound begin to proliferate and form new capillaries that gradually migrate into the wound. The newly developing capillaries

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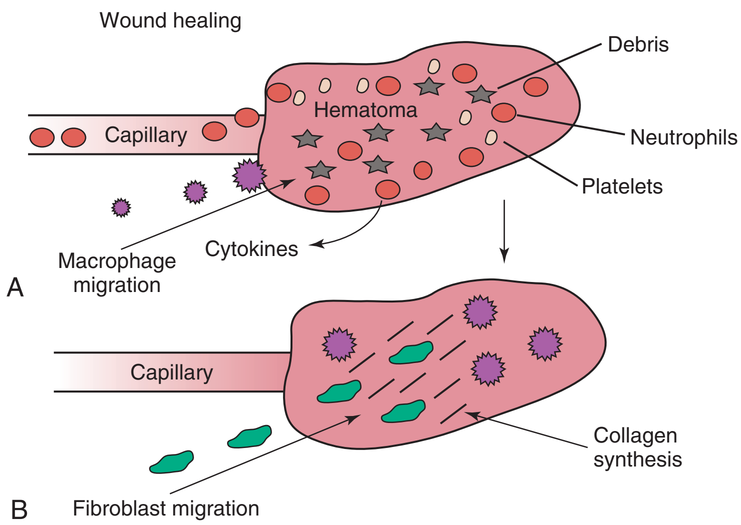

Figure 6-1 Inflammatory (*A*) and reparative (*B*) phases of wound healing.

Box 6-1 Healing Timeline: General

Inflammation
Immediate to 3-5 days

Repair
72 hr to 2 wk

Remodeling/Maturation
Occurs in 2-3 wk (max collagen deposition) to 1 yr

Healing/Strength Characteristics
Varies

Implications for Rehabilitation
N/A
Rates are generalizations of individual tissue healing. Results from each injury vary depending on the severity, chronicity, and age of the individual.

Box 6-2 Healing Timeline: Skin

Inflammation
N/A

Repair
Collagen stabilizes at approximately 21 days

Remodeling/Maturation
N/A

Healing/Strength Characteristics
Occurs in 21 days (20% normal strength) to 1 yr (70% normal strength)

Implications for Rehabilitation
Minimal effect

follow immediately behind the migrating fibroblast and collagen scaffold and continue until normal oxygen tension in the wound is restored.[5] This dense network of macrophages, fibroblasts, and neovascularization during the proliferative phase is generally recognized as granulation tissue.

Remodeling (Maturation) Phase

The remodeling phase is the final aspect of wound healing during which collagen fibers reorient parallel to lines of stress and strain and the fibers cross link in a stable formation.[7,8] This is the most important aspect of wound healing for connective tissues, because appropriate collagen deposition and alignment are critical to adequate tensile strength development in the repaired tissue. Although collagen deposition reaches a maximal point 2 to 3 weeks after injury, tensile strength continues to progressively increase over the course of approximately 1 year (Box 6-1 and Box 6-2). This period allows for removal of biomechanically inferior collagen fibers (type III) and replacement with the fibers suitable for the specific tissue involved (generally type I collagen). In addition, decreasing proteoglycan concentration leads to a decrease in water content, resulting in compression of the collagen fibers. As fibers become closer in proximity because of realignment and compression, increased surface area is available for cross-linking, subsequently increasing tensile strength.[5] Thus a proper equilibrium between collagenolysis and matrix accumulation becomes essential.

Bone Healing

Bone Structure

Bone is a dynamic tissue composed of 35% organic material and 65% mineral.[2,9-11] The organic portion comprises cellular constituents (osteocytes, osteoblasts, and osteoclasts) and matrix components. The osteocytes are cells of mature bone that remain within spaces (lacunae) surrounded by lamellar bone and are responsible for normal mineral homeostasis. Osteoblasts are responsible for osteoid production, and osteoclasts function in bone resorption.[11,12] A fine balance of bone resorption and deposition is maintained for mineral homeostasis as well as continuous remodeling secondary to mechanical stresses placed on the bone in accordance with Wolff's law which states that a bone in a healthy individual will adapt to the forces under which it is placed.[12,13]

Bone matrix is composed of predominately type I collagen and ground substance containing glycosaminoglycans.[8,11,12] Calcium and phosphorus deposit in association with the collagen fibrils and serve as the primary reservoir for overall calcium homeostasis. Hydroxyapatite crystals of calcium and phosphorus are aligned with the collagen fibrils and provide the structural rigidity to bone.[11,12] This unique arrangement between collagen and mineral creates the viscoelastic property of bone with the crystal supplying resistance to compressive forces and the collagen providing tensile support.[12]

Blood supply to bone is primarily based on the nutrient artery that bifurcates into ascending and descending branches within the medullary cavity (Figure 6-2). In addition, there is vascular support from metaphyseal arteries and periosteal arterioles.[11] The metaphyseal vessels anastomose with the nutrient artery branches within the medullary canal to provide complete collateral circulation. The vascular flow is centrifugal, with elevated pressure in the medullary cavity driving blood toward the periosteum. The efferent system is a venous drainage system using periosteal venules and the medullary venous sinus. Cortical vessels flow through Haversian canals in which small vascular channels are surrounded by layers of lamellar bone.[9-11] These channels in the cortical bone provide perfusion to the osteocytes within their lacunae.[12] The periosteal arterioles are the least important contributors of the vascular supply in healthy intact bone and are primarily of concern in areas of soft tissue attachment. Injury to the bone results in disruption of the vascular supply to varying degrees depending on the severity of the fracture. If the endosteal supply remains intact, it will predominate during healing. However, if the endosteal supply is disrupted, as in most fractures, the periosteal support becomes extremely important, thus making careful maintenance and handling of soft tissue attachments crucial to healing (Box 6-3).[14]

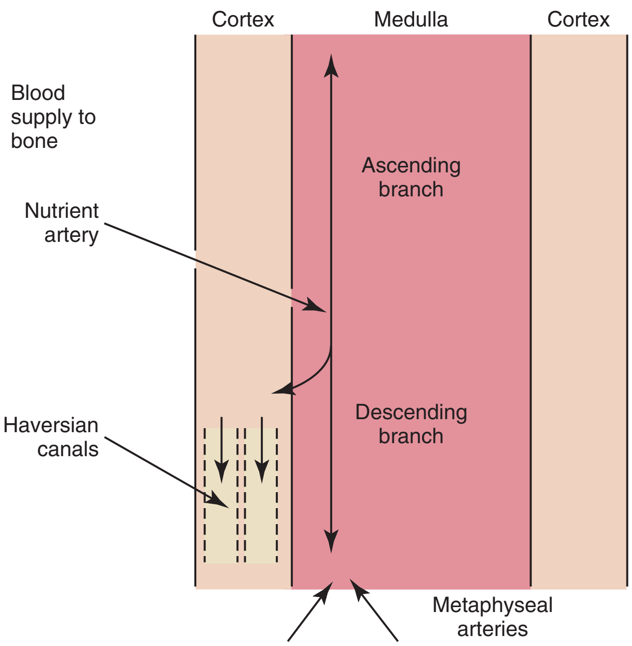

Figure 6-2 Blood supply to bone.

Primary Bone Healing

As discussed previously, bone is one of the few tissues that can undergo direct cellular regeneration to restore 100% of the original biomechanical properties. This can occur through two major types of healing, primary or secondary. Primary, or direct, bone healing occurs when there is a minimal fracture gap with rigid stability, and it proceeds by either contact or gap healing. Contact healing involves the direct formation of bone across a fracture line less than 0.1 mm wide by creating resorption cavities, or "cutting cones," through the Haversian canals parallel to the long axis of the bone (Figure 6-3).[9,10,15,16] Osteoclasts lead the front edge of the resorption cavity across the fracture line and remove bone and calcified matrix in preparation for osteoblastic activity immediately following. Finally, nutritional support is supplied by capillary loops following the advancing osteoblasts.[9,15,16] This progressive Haversian remodeling continues across the fracture line until the surrounding bone has been remodeled into new Haversian systems.

When the fracture line is greater than 0.1 mm but less than 0.5 mm, primary bone union can still occur through gap healing.[10,16] This distance cannot be directly crossed by a resorption cavity. As a result, lamellar bone produced from cells lining the medullary cavity and periosteum first fills the void in a transverse direction. Subsequently, resorption cavities cross the newly formed lamellar bone and again create new Haversian systems in the proper longitudinal direction.[10,16] In fracture repair, both gap and contact healing may occur in different portions of the same fracture line.[16] With primary bone healing, there is little radiographic callus formation, and the fracture line gradually disappears. Although direct healing entails direct formation of lamellar bone and remodeling of Haversian systems without intermediate fibrous or cartilaginous tissue, it does not reduce the time for fracture healing, and weeks to months may still be required before complete union.[15] In fact, early stability is generally decreased when compared with secondary healing, typically requiring longer periods before removal of stabilizing implants.[17] Radiographic evaluation is generally performed at 4- to

Box 6-3 Healing Timeline: Bone

Inflammation

Primary and secondary bone healing follow similar timelines. Immediate to 72 hr (hematoma and inflammation, secondary bone healing)

Repair

Radiographic healing (secondary): appearance of callus at 10-12 days, disappearance of fracture line within 30 days.[85] Disappearance of fracture line radiographically as early as 30 days with secondary bone healing. Primary bone healing takes longer.

Remodeling/Maturation

Woven bone development over mineralized callus. Complete remodeling of the callus 90 days after fracture repair for a simple fracture.[85] Primary bone healing (direct union, no callus formation) produces less early stability than secondary, and requires implants to be in place for a longer time than with secondary healing.[17]

Healing/Strength Characteristics

Regaining 100% tensile strength is possible because of regeneration. Healing and strength are faster with secondary bone healing and callous formation compared with primary bone healing. Rate of healing depends on presence of interfragmentary strain (faster at less than 2%), animal age, nature and location of the fracture (faster in sites with abundant cancellous bone and highly vascularized marrow), presence or absence of concurrent disease and surgical fixation method.[85] *Clinical union* defined as ability to withstand load so that fixation can be removed. In general, bone achieves clinical union according to the following timelines[17]:

- Puppies under 3 months: 2-4 wk
- Puppies 3-6 months: 4-12 wk
- Puppies 6-12 months: 5-8 wk to 3-5 mo
- Adult dogs (1 year of age or older): 7-12 wk to 5-12 mo

Rate is faster for IM pins and external skeletal fixation than for bone plating. Destabilize external fixators at approximately 4-6 wk in dogs.[20, 21]

Implications for Rehabilitation

Rehabilitation goals throughout healing have been suggested.[86] During inflammatory phase (immediately up to 2-3 days after injury), decrease pain, minimize soft tissue inflammation, maintain joint health/Range of motion (ROM) and establish safe, controlled mobility, depending on fracture stability. During repair phase (3 days to 4-6 wk after injury), small, gradually increasing cyclic loading to increase micromotion and improve callus formation with secondary healing; continued maintenance of joint health through ROM, active or passive.

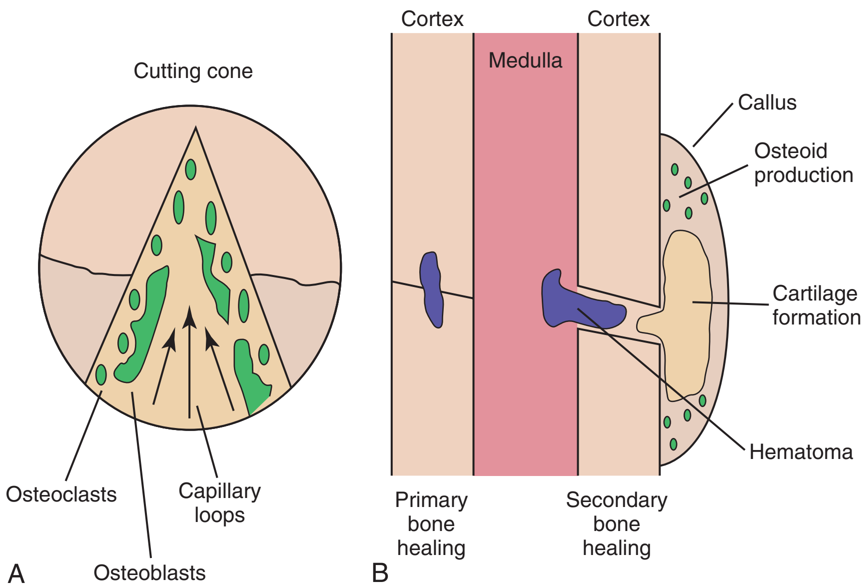

Figure 6-3 Bone healing. **A,** Primary bone healing with the formation of cutting cones. **B,** Secondary bone healing.

6-week intervals until the fracture line has resolved. The goal is to achieve recovery of function and early weight-bearing with implants employing interfragmentary compression so rehabilitation can begin at an early stage.

Secondary Bone Healing

Secondary, or indirect, bone healing occurs when there is less rigidity or absence of anatomic reduction such that the fracture gap is greater than 0.5 mm and is characterized by callus formation (see Figure 6-3).[16,18] The degree of callus formation is related to the amount of instability at the fracture site. In general, secondary bone healing proceeds through the same phases as basic wound healing.[9,10,19]

Similar to basic wound healing, hemostasis is the initial response at the time of injury, initiating the inflammatory phase. Immediate hemorrhage and hematoma formation occur within the fracture gap in addition to the surrounding soft tissues that may be damaged. Destruction of the Haversian canals at the site of injury causes osteocyte death and necrosis of bone. The process proceeds with the migration of neutrophils followed by macrophages into the injured site for removal of necrotic debris. In addition, biochemical mediators such as bone morphogenetic proteins, platelet-derived growth factor, osteonectin, prostaglandins, and osteocalcin enter the fracture site to contribute to bone healing.[10] As the inflammatory response previously described subsides (24 to 72 hours), the repair process begins.[9,10,19]

The reparative phase begins with the surrounding tissue providing mesenchymal cells, which enter the fracture site to form osteoblasts, chondroblasts, and fibroblasts. During the initial 3 weeks, a fibrous and cartilaginous "soft" callus is formed by the deposition of types I, II, III, V, IX, and X collagen synthesized by mesenchymal cells. Types III and V collagen initially appear during the inflammatory phase, whereas types II and IX collagen peak during the cartilaginous aspect of the reparative phase 1 to 2 weeks after injury. Collagen type I is initially present at low levels in the first 2 weeks but becomes the predominant type by day 14.[8,13] The function of the soft callus is to unite the fracture ends and decrease interfragmentary motion and strain.[16] The fibrocartilage deposited is gradually transformed to bone by the process of endochondral ossification similar to that which occurs during physeal growth.[8,19] Chondrocytes progress to a hypertrophied stage followed by mineralization of the surrounding matrix. Invasion by capillaries brings cells that resorb the mineralized cartilage matrix, produce osteoid, and deposit bone. This transition from soft cartilaginous callus to hard bony callus is directly related to the blood supply available and the interfragmentary strain at the fracture gap. With high interfragmentary strain or poor blood supply, activity of osteoblasts and osteoid production is limited, and the callus remains as fibrocartilage.[10,16]

The remodeling or maturation phase proceeds following the transition from soft to hard callus. The fracture gap is bridged and stabilized with endosteal and periosteal callus consisting primarily of woven bone that must be remodeled to lamellar bone according to weight-bearing forces as directed by Wolff's law. In addition, the normal blood supply is reestablished. As the endosteal callus is gradually remodeled, the medullary cavity and its blood supply are restored. Similarly, as cortical bone is restored, the Haversian system is re-created and the normal centrifugal vascular pattern can return with concurrent resorption of the temporary extraosseous supply.[18,19]

Factors Affecting Healing

Biomechanical factors at the fracture site are important to healing. Wolff's law has a tremendous effect on bone deposition and resorption. The stress placed on the bone stimulates healing as long as the micromotion created does not exceed the acceptable interfragmentary strain for osteoid deposition.[18] If the mechanical load exceeds the strength of the reparative tissues, inhibition of bone healing or re-fracture occurs. Thus it is important that no or minimal motion between fracture fragments occurs early in the healing phase to allow adequate establishment of the extraosseous blood supply. After the vascular supply is established and callus formation has occurred, then controlled axial micromotion can stimulate appropriate bone remodeling. As such, destabilization of external fixator frames has been recommended at approximately 4 to 6 weeks following frame application to "dynamize" the fracture line.[20,21]

Electrical fields have been demonstrated in normal and injured bone and are related to either mechanical forces or bioelectric potentials.[22] Strain-related potentials occur secondary to mechanical loading of the bone and are related to piezoelectric activity resulting from collagen deformation and fluid flowing through space in the extracellular matrix.[10] Bioelectric potentials are generated by cellular functions associated with metabolic processes in the cell.[22] These electrical fields can be stimulated exogenously and have been demonstrated to favorably affect bone remodeling. Electrical stimulation can be provided by invasive or noninvasive methods. Invasive stimulation includes direct surgical implantation of a cathode at the site of stimulation and the anode placed in the surrounding soft tissue. Noninvasive stimulation can be accomplished by capacitive or inductive coupling of current to the appropriate site. With capacitive coupling, the electrodes are placed on the skin on opposite sides of the stimulation site, and electrical fields between 1 and 10 mV/cm are generated. Inductive coupling uses external electromagnetic coils to establish an electromagnetic field of time-varying or pulsed nature to create a secondary electrical field in the bone.[22] Currently, the mechanism of electrical stimulation of bone healing is not completely understood, but its use has been advocated with nonunion fractures.[23,24]

Alterations in endocrine balance can influence bone healing as well. Hormones such as parathormone, calcitonin, vitamin D, and thyroid may affect bone resorption and deposition.[10] Corticosteroids and diabetes mellitus can have inhibitory effects on bone healing.[10,25] In addition, there are numerous other miscellaneous factors that must be considered with respect to fracture healing, including patient age, fracture configuration, fracture location, and whether the site was grafted.[10,16] Young patients heal faster than geriatric individuals. The fracture configuration is also important because highly comminuted fractures or those with significant bone loss require more time to achieve bony union. Furthermore, fractures closer to the metaphysis in locations of high cancellous bone, or those that have been grafted, may have more rapid healing potential.

Efforts to enhance fracture healing continue along many pathways. The ability to stimulate bone healing with the use of growth factors is a future possibility, and research continues in the field of electrical stimulation. In addition, research has demonstrated potential advantages to using ultrasound and extracorporeal shockwave technology to enhance fracture healing.[24,26]

Muscle Healing

Muscle Structure

Skeletal muscle is composed of aggregations of muscle cells termed *fibers* (*myofibers*), consisting of multiple nuclei within each fiber (Figure 6-4). Each individual muscle belly is composed of numerous fibers bound together by connective tissue into multiple diffuse bundles (fascicles). A dense fibrous connective tissue sheath termed the *epimysium* surrounds the entire muscle. The *perimysium* is the connective tissue support surrounding each fiber bundle within the muscle belly and the *endomysium* represents sheets of connective tissue surrounding each independent muscle fiber.[27,28] Capillaries, nerves, and lymphatics traverse to the muscle fibers through the surrounding connective tissue sheaths, eventually reaching the endomysium, where a rich capillary network highly sensitive to sympathetic vasomotor tone is established around the individual myofibers.[29]

Each muscle fiber has an extensive subcellular structure. Myofibrils within the muscle fiber are composed of both thick (myosin) and thin (actin) filaments, troponin, and tropomyosin, which function to provide the contractile properties of the muscle fiber.[27,29] The myofibrils are surrounded closely by the sarcoplasmic reticulum, which plays a role in storage and release of calcium for muscle function.[29] Variations in the composition of the myofibers and sarcoplasmic reticulum result in different types of fibers (slow or fast twitch) for various muscular functions and fatigue resistance.

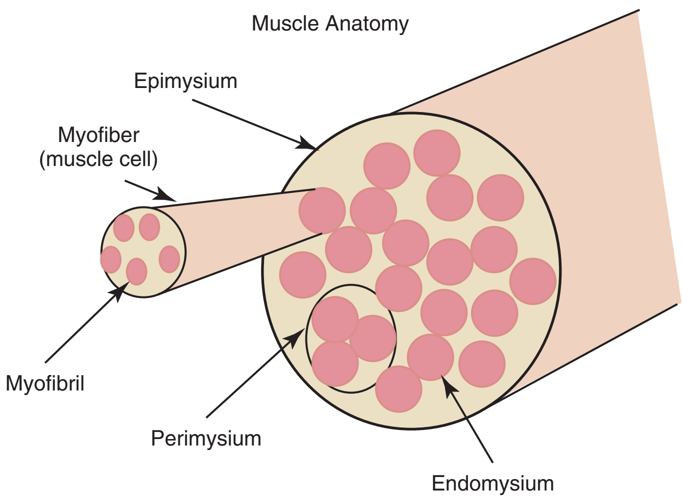

Figure 6-4 Anatomy of muscle connective tissue.

Muscle Injury and Healing

Injury to muscle occurs as a result of lacerations, contusions, ruptures, ischemia, and strains. A *strain* is described as an injury to the muscle or tendon with a tendency to occur near the myotendinous junction. Strains are described as grade 1 through 4, with grade 1 strains being disruption of a few muscle fibers and grade 4 being complete rupture of the muscle belly.[30] Often the cause of nonpenetrating injuries is forceful contraction of the muscle belly with concurrent passive extension.[28] Damage to the muscle results in tearing of the muscle fibers and disruption of the vascular or connective tissue support. The problem may be acute or chronic, and the severity may vary from mild injury to complete rupture.

Muscle healing progresses in the same basic pattern as general wound healing (Box 6-4). An initial period of hemostasis and inflammation precedes repair. Following

Box 6-4 Healing Timeline: Skeletal Muscle

Inflammation

Immediate to 72 hr (inflammation)

Repair

3 days (type III collagen) for scaffold for myofibers; normal collagen ratio > 6 wk

Remodeling/Maturation

Ratio of functional muscle fiber to scar tissue depends greatly on size of gap at site of injury

Healing/Strength Characteristics

Normal tissue regeneration is required for return to full function; occurs in minimum 4-6 wk.[28] Complete muscle lacerations near midbelly in rabbits showed final return to 50% of tension-producing ability and 80% of normal contraction shortening at 12 wk.[71] Partial tears have an improved outcome for tensile strength; return to 60% of tensile strength and normal contractile ability achieved with partial lacerations at 12 wk.[71]

Implications for Rehabilitation

Decreases in muscle contraction strength directly related to degree of injury and ingrowth of fibrous tissue. Rehabilitation depends on extent of injury (i.e., degree of strain, rupture). In mild strains, brief (3-5 day) immobilization followed by active muscle contractions in mobilization demonstrates better penetration of muscle fibers through the scar tissue and better muscle strength and size.[72] For muscle tears, immobilize for a minimum of 2-3 wk to protect repair, promote new collagen, and sustain remobilization stresses.[33,34] For rehabilitation of complete rupture, moderate activity can begin at 4-8 wk. At 8-12 wk, unrestricted activity may be allowed.[73]

hematoma formation, the edema and ischemia can result in necrosis of adjacent ruptured muscle fibers.[31] The initial inflammatory phase continues with neutrophil and macrophage removal of necrotic debris during the first 24 to 72 hours.[28,30] A distinct relationship between macrophage penetration and myofiber regeneration has been suspected; however, the essential component is reestablishment of the vascular supply to the injured fibers.[29] The repair phase follows and is a competitive process involving either regeneration of functional myofibers or production of fibrous scar tissue, depending on the severity of the injury and the size of muscle gap.[30] The repair by scar tissue production is not as desirable because of the higher chance of recurrent injury and a decrease in muscle contractile strength of approximately 50%.[30,32]

As in general wound healing, the repair phase continues with production of extracellular matrix by fibroblasts including proteoglycans, fibronectin, and collagen. Type III collagen appears within the injured site in the first 3 days and generally precedes the thicker type I collagen fiber production.[33] This extracellular matrix provides a scaffold for myofiber regeneration across the injury site.[30] Myoblastic stem cells from the surrounding wound environment (satellite cells) and mononuclear leukocytes migrate to the viable ends of myofibers and align across the gap, providing a route for myotubules to cross the injured site and repair the myofiber.[30] The alternative form of healing occurs when there is an inadequate source of myoblasts, inadequate vascularization, inadequate innervation, or excessive stress across the injured site creating a significant gap.[28,30] In this situation, excessive collagen deposition and fibroblastic activity lead to formation of fibrous tissue within the injured site. The result is a barrier to myofibers crossing the injury site and inadequate return to normal muscle function.

Timing of appropriate stress and motion at the injury site is important to the type of healing that occurs. For optimal muscle fiber healing and return to maximal function, mobility across the injury site should not begin until healing is well under way, such as in the reparative or early remodeling phases. During the reparative and remodeling phases, the muscle fibers can be aligned parallel to the line of action if appropriate stresses are allowed.[30] If motion is excessive, gap widening occurs and the muscle heals with more fibrous tissue.

Prolonged immobilization allows penetration of regenerating myofibers and decreased fibrous tissue formation; however, the muscle fibers tend to orient inappropriately to lines of stress and tension. This decreases the tensile strength of the muscle fibers, and if the fibrous tissue elongates during the remodeling phase, significant loss of muscle power can occur.[28,30] An initial immobilization period of 3 to 5 days followed by controlled mobilization accelerates the appearance of type I collagen fiber production and subsequently increases tensile strength.[33,34] Considering the normal timing associated with wound healing, recommendations have been made for 4 to 6 weeks of protected activity before increased activity.[28] Adequate approximation of the myofiber ends and controlled mobilization are extremely important to adequate long-term muscle healing and function.

Tendon Healing

Tendon Structure

Tendons, similar to ligaments, are composed primarily of type I collagen (>95%) organized into parallel bundles with surrounding extracellular matrix. Fibroblasts or tenocytes are the predominant cellular component of the tissue and function in repair and maintenance of the structure. The individual collagen bundles are surrounded by a loose connective tissue termed *endotenon*. The epitenon covers the entire group of bundles, and the paratenon covers and separates different tendon bellies, providing the potential for smooth gliding function.[28,35] Tendons are classified as either *vascular* (lacking synovial sheaths) or *avascular* (encased within a synovial lining). Tendons within synovial sheaths lack the typical paratenon. The blood supply to the tendon arises from three primary sources: the musculotendinous junction, the osseous point of insertion, and adjacent muscle and paratenon.[28]

Tendon Healing

The process of tendon healing follows the general course of wound healing as previously discussed with initial fibroplasia and collagen deposition followed by maturation and reorientation of the collagen fibrils (Box 6-5). The primary variable in tendon healing is whether the tendon is a vascular paratenon-covered tendon or a synovial-sheathed tendon. Paratenon-covered tendons rely less heavily on the intrinsic blood supply, as fibroblasts and capillaries invade the injured tendon ends from the surrounding tissue. This establishes the “one wound–one scar” theory.[35] With synovial-lined tendons, a longer period is required to achieve adequate tensile strength and healing because healing depends on the tendon’s intrinsic blood supply or inflammatory repair tissue infiltration. Thus immobilization is extremely important in the initial period to resist gap formation at the repair site and allow adequate healing.[30,36,37]

Early descriptions of tendon healing involved an extrinsic process in which an influx of cells from the synovial sheath, paratenon, or epitenon allowed phagocytosis, fibroplasia, and subsequent collagen deposition.[28,38,39] The inflammatory stage is seen within the first 3 days and is quickly followed by randomly oriented collagen deposition by the fifth day. The reparative phase follows with progressively increasing collagen content through the first 4 weeks. Similar to muscle healing, the initial collagen is

Box 6-5 Healing Timeline: Tendon

Inflammation

Immediate to 72 hr

Repair

Day 5: random collagen orientation, progressively increasing for 4 wk; collagen production peaks 5-12 days after injury, markedly decreased by day 60

Remodeling/Maturation

Takes a minimum 28 days for collagen fibers to align according to stress; tendons appear normal by 112 days

Healing/Strength Characteristics

At 6 wk, 56% normal tensile strength occurs, 79% at 1 year and beyond.[36] Surgical repair of fresh gap required for appropriate organization of scar at 6 wk. Tendons with gaps > 3 mm have lower tensile strength at 6 wk compared to those with smaller gaps.[74] Synovial (avascular) tendons require longer healing times than vascular tendons.

Implications for Rehabilitation

Post-repair support for larger tendons may be needed for up to 6 wk. Method and extent varies depending on surgeon preference, location of the injury and patient behavior.[34] Prolonged immobilization may be detrimental. Complete immobilization for longer than 21 days results in reduced vascularization at the wound site.[75] With full or partial transection and surgical repair, passive mobilization exercises appear to be optimal to begin at day 5 following surgery.[76] Early, controlled passive mobilization continuing for 21 days appears to augment intrinsic healing, increase tensile strength and reduce adhesion formation.[41-46] One study showed active mobilization (cage activity) was not detrimental at 3 wk after injury for repaired partial flexor tendon lacerations up to 60%.[77] Conservative exercise with light weightbearing may begin at 6 wk, because having 25-50% of normal tensile strength for tendons has been considered adequate for withstanding normal muscle force. Minimum 12 wk for less restrained, more active loading.[78, 79] Light exercise to maintain ROM and stimulate healing callus cells. Some weightbearing is required for maturation of callus. Sufficient tensile strength return for normal activity load bearing may take months.

generally type III, which is replaced with the biomechanically superior type I collagen as repair and remodeling progress.[8,40] Research has demonstrated that healing of tendon ends by collagen deposition requires a minimum of 28 days for fibers to align parallel to lines of stress. In addition, collagen bundles are distinguishable from normal tendons until 112 days after injury.[38] Evaluation of healing of canine triceps tenotomies (a vascular tendon lacking a synovial sheath) demonstrated only 56% of breaking strength at 6 weeks and only 79% tensile strength 1 year following injury.[36]

More recently, evidence has been presented of the potential for intrinsic repair controlled predominately by the epitenon and endotenon.[41-44] This repair mechanism occurs primarily in synovial sheathed tendons with minimal damage to the synovial lining. If there is severe damage to the synovial sheath, the extrinsic mechanism overwhelms the capacity for intrinsic healing. The intrinsic pattern of cellular proliferation and sequencing is similar to extrinsic healing; however, there is a balance between epitenon and endotenon contributions and there is minimal adhesion formation.[40] The intrinsic healing process can be stimulated by early controlled mobilization, decreasing the risk of adhesion formation and providing greater tensile strength as a result of proper alignment of collagen fibers along the lines of stress.[41-46] Early controlled passive motion beginning within 21 days of repair appears to be ideal; however, the duration and magnitude of motion for optimal healing remain unknown. In addition, patient compliance in veterinary medicine is a problem and must be considered to ensure a good outcome.[30,42]

Tendon Repair

The goal of treatment is to minimize adhesion formation and return to maximum function. The appropriate treatment is surgical apposition of the transected or injured tendon ends. Numerous suture patterns have been designed to maximize tensile strength and attempt to minimize gap formation at the injury site. Commonly used techniques include the locking-loop and three-loop pulley patterns.[35,47-52] Adherence to several fundamental concepts is required. Meticulous hemostasis is necessary to minimize hematoma formation within the injury site and minimize adhesion formation. In addition, atraumatic technique and minimal handling of the tendon ends is important. Suture materials should be monofilament to minimize reactivity and nonabsorbable to allow time for the return of adequate tensile strength. Monofilament suture also allows efficient gliding through the tissue and even distribution of weightbearing forces and tendon tension.[28,35]

Ligament Healing

Ligament Structure

Ligaments are dense connective tissue structures consisting of fibroblasts, water, collagen, proteoglycans, fibronectin, and elastin that connect two or more bones.[11,53] They tend to widen at insertion points to blend with the periosteum and generally provide support to the articulating surfaces. Ligaments are primarily composed of long parallel or spiral collagenous fibers that are strategically arranged in multiple directions to restrain the joint from excessive excursions secondary to both normal and abnormal motions.[8,11] Histologically, the fibroblasts have elongated basophilic nuclei lying between collagen bundles.[53] The cellular and collagenous densities vary with each ligament. Collagen makes up 70% to 80% of ligament dry weight composition, with the majority being type I fibers and approximately 10% being type III fibers.[53] The collagen fibers are generally crimped or buckled and can be stretched out with stress to provide elasticity and the potential to withstand rapid load application.[53]

Ligament Healing

Currently, a great deal of information remains unanswered regarding timing of ligamentous healing, especially with respect to postoperative mobilization techniques. This is because ligaments heal differently depending on the location. For example, the healing potential of the medial collateral ligament of the stifle is very good,[53,54] but the cranial cruciate ligament, which has received the most investigation, demonstrates virtually no healing response following injury.[55] However, it appears that ligaments generally heal according to the same basic pattern of inflammation, repair, and remodeling (Box 6-6).[55,56]

Within hours of injury, the defect is filled with an organized hematoma and the surrounding tissue becomes edematous from perivascular leakage of fluid. This inflammatory stage is characterized by an influx of inflammatory cells as in general wound healing. Monocytes and macrophages are found in the wound by 24 hours and respond by cleaning up the site and transitioning to the next phase. This stage lasts for approximately 48 to 72 hours.[55]

The reparative phase begins 2 to 3 days after injury and persists for approximately 6 weeks.[55] Both cellular and matrix production predominate, with granulation tissue and vascular ingrowth filling the defect between friable ligament ends. The scar in the first few weeks is highly cellular with fibroblasts actively synthesizing primarily types I and III collagen and other extracellular matrix components.[55] During this aspect of ligament healing, the scar cross-sectional area reaches a maximum.[30] Type I collagen predominates and tensile strength of the ligamentous repair tissue increases as time progresses.

The final stage of remodeling and maturation progresses as with normal wound healing and may take more than 12 months to complete. At that point, ligament strength is only 50% to 70% of the original tensile strength.[55] Part of the lost strength may also be a result of bone loss at the ligament insertion sites. In general, healing of the midsubstance of the ligament progresses more rapidly than the insertion sites because bone turnover and recovery are slower. Fibroblast and macrophage numbers decrease, and the collagen fibers become more densely packed and appropriately aligned. Factors that are important to appropriate repair of tissue include ligament end apposition, nutritional status, endocrine imbalances, severity of injury, blood supply, and mechanical stresses placed across the healing tissue.[30,55] As with muscle healing, allowing the ligament to heal across a large gap results in excessive scar tissue formation that obstructs histologically normal ligament from bridging the defect.[30]

Box 6-6 Healing Timeline: Ligament (Sprain)

Inflammation

Immediate to 72 hr

Repair

2-3 days to 6 wk

Remodeling/Maturation

N/A

Healing/Strength Characteristics

Occurs in several days to more than 1 yr, depending on grade (I-III). At 1 yr, 50-70% tensile strength occurs.[65] Healing properties and rate of healing depend on which ligaments are involved.[80,81] Transected canine stifle medial collateral ligament (MCL) strength 52% of controls after 14 wk of healing, reduced to 14% of controls when cranial cruciate ligament (CCL) also transected. Transected canine CCL weakness persists after 4 yr of healing (hence other methods of surgical repair needed). Lengthening may occur because of gap formation.

Implications for Rehabilitation

Time to begin rehabilitation varies widely depending on the ligament involved. When in doubt, parameters for tendon healing may generally be followed. Tensile strength reduction may be slowed or decreased with joint motion during immobilization and stress reduction. No change at 6 wk and 67% reduction at 12 wk in repositioned canine CCL in one study.[82] Low-duration (30 min) and high frequency (6 days/wk) exercise appears to have greatest benefit to ligament strength from one study in horses.[83] Sprain-avulsion fractures at ligament-bone interface have best prognosis for return to normal tensile strength; fractures heal within 2-4 mo, which restores ligament integrity.[84]

Articular Cartilage Healing

Articular Cartilage Structure and Function

Diarthrodial (synovial) joints consist of two or more hyaline cartilage surfaces that are tightly adhered to the underlying cortical end plates of apposing bones. Hyaline cartilage is designed to provide a smooth, low-friction, gliding surface for joint movement and to transmit weight-bearing forces to the underlying bone structure.[11,57-60] Cartilage is primarily avascular and without lymphatics. It is approximately 70% to 80% water by weight with the remainder composed of chondrocytes, type II collagen (90% to 95% of the total cartilage collagen), and proteoglycan aggregates.[11,57-61] Collagen provides tensile strength to the joint surface, which is divided into three zones. The most superficial layer, the tangential zone, has collagen fibers aligned parallel to the joint surface, creating primary resistance to shear and tensile stresses placed on the cartilage surface. Deeper within the cartilage, the collagen fibers become oriented in a more oblique fashion and form the intermediate zone. The final, and largest, zone is the radial zone in which the collagen fibers are oriented in a perpendicular columnar arrangement and become embedded into the subchondral bone.[57-61]

Chondrocytes compose less than 10% of the total cartilage volume.[61] The chondrocytes are oriented similar to collagen except for an additional zone of calcification present deep within the cartilage to provide transition to the subchondral bone. Surrounding each chondrocyte is the cartilage matrix separated into three regions classified as pericellular, territorial, and interterritorial. The function of the chondrocyte early in life is primarily cell proliferation and matrix synthesis. As maturation proceeds, these processes slow, with cellular density decreasing and interterritorial matrix size increasing.[57-61]

Proteoglycan aggregates provide the compressive strength to cartilage and progressively increase in quantity from the intermediate to the radial zones of the cartilage.[58] The primary constituents of the proteoglycan aggregate are a series of glycosaminoglycans attached to a core protein, which in turn are linked to a backbone chain of hyaluronic acid. The glycosaminoglycans are composed of chondroitin-6 sulfate, keratan sulfate, and some chondroitin-4 sulfate. These highly sulfated and carboxylated structures carry negative charges that strongly repel each other, resulting in stiffly extended aggregates. In addition, they are highly hydrophilic. The resulting osmotic swelling pressure, along with the fixed anionic charges, produces dramatic compressive resistance. These products are synthesized by chondrocytes and provide supportive structure to the collagen fiber network.[56-61]

A capsule surrounds the joint and is created by an outer fibrous layer and an inner synovial membrane. The outer fibrous layer, continuous with the periosteum, is composed primarily of collagen and helps provide mechanical support. The inner synovial membrane is a thin layer of synovial cells closely associated with a subsynovial vascular plexus.[11,57-59] The cellular component of the synovial membrane is divided into type A synoviocytes, which are primarily responsible for phagocytic activity, and type B synoviocytes, which are responsible for hyaluronic acid production.[57,58] Synovial fluid is a plasma ultrafiltrate from the subsynovial plexus, which allows molecules less than 12,000 daltons to permeate into the joint. To this ultrafiltrate, the type B synoviocytes add hyaluronic acid.[57,58] Synovial fluid provides joint lubrication and cartilage nutrition, which occur through two separate processes.[11,57,58] Boundary lubrication is associated with the synovial membrane, the periphery of the cartilage surface, and hyaluronic acid. Hydrostatic lubrication is a process created by the weight-bearing force on the cartilage surface, which pushes water to the surface during weight bearing. Weight bearing and joint motion also improve the diffusion of nutrients to the chondrocytes and the removal of waste metabolites.

Cartilage Healing

Injury to articular cartilage and the subsequent healing response are directly related to the type of trauma incurred. Similar to the basic pattern of general wound healing, the ideal response includes initial necrosis and inflammation, followed by vascular ingrowth and the reparative phase (Box 6-7). The limiting factor for the repair of articular cartilage is its avascular nature. As a result, injury to the articular surface results in two separate responses based on whether the injury is limited to the cartilage surface or extends into the subchondral bone.[62,63]

Box 6-7 Tissue Healing: Articular Cartilage

Partial Thickness

Inflammation
No inflammatory response—avascular

Repair
Mitotic activity ceases 1 wk after injury

Remodeling/Maturation
Local chondrocyte activity does not adequately fill the defect at 1 yr because of lack of inflammatory healing response

Healing/Strength Characteristics
N/A

Implications for Rehabilitation
N/A

Full Thickness (to subchondral bone)

Inflammation
Immediate to 3-5 days (inflammatory response from marrow cells)

Repair
5-7 days to 2 mo

Remodeling/Maturation
Occurs in 2-6 mo; by 6 mo repair tissue changed significantly from normal hyaline cartilage. Repaired tissue is fibrous and inferior to hyaline cartilage. At 1 yr, 20% is type II collagen.[63]

Healing/Strength Characteristics
Defect repair depends on patient age, size, and location of the defect. Regardless, complete restoration of hyaline articular cartilage is very rare.[87]

Implications for Rehabilitation
Continuous passive range of motion (PROM) after injury appears to improve healing with mainly hyaline cartilage observed in 60% of defects (adolescent) and 44% of defects (adult) in rabbits undergoing continuous PROM, compared to 10% or fewer in controls (immobilization in a cast or free-range in cages).[87] Less profound results occur in defects larger than 3 mm in diameter.

Injury limited to the cartilage surface results in a number of secondary responses. A traumatic laceration perpendicular to the joint surface (without invasion below the calcified cartilage layer) results in localized death of the chondrocytes and subsequent loss of matrix support. Consequently, a matrix defect occurs that is characterized by a lack of vascular infiltration and no inflammatory response. As a result, there is no fibroblastic response, and the local chondrocytes must proliferate and fill the defect with new matrix.[62,63] The chondrocytes respond with an initial increase in mitotic activity adjacent to the defect.[64] However, the chondrocytes may not completely fill the defect and a lesion often remains. With a lesion parallel to the cartilage surface, a similar response ensues with initial limited cellular proliferation and increased proteoglycan and matrix synthesis to fill the injured site. The primary problem is the limited durability of the repair process and the failure to completely replace the cartilage matrix. If the defect is large, then degenerative joint disease may ensue. It appears that if the defect is small, such as in a surgical laceration, it will generally be limited in progression and not proceed to significant degenerative change within the joint.[62,63,65]

Full-thickness injury to the level of the subchondral bone allows access of marrow stem cells and vascular capillaries to the cartilage defect. As a result, the injury can respond similar to general wound repair with an initial debridement and inflammatory phase. Within 48 hours a fibrin clot forms, providing a scaffold for migration of fibroblasts.[63,65] Fibroblasts and collagen replace the clot in 5 to 7 days and metaplasia to fibrocartilage begins. Matrix formation occurs with a proteoglycan-rich environment, and defect repair takes approximately 2 months to complete.[63,65] Wound healing continues with remodeling and maturation. At 6 months, the glycosaminoglycan concentration decreases significantly and tends to be rich in dermatan sulfate, which is smaller and less wear resistant than other glycosaminoglycans. Type I collagen predominates in the early stages but slowly decreases with time (40% at 8 weeks, 20% at 1 year).[63] The repair tissue remains more fibrous and is biomechanically inferior to the normal surrounding hyaline cartilage with a gross distinction permanently remaining.[62,63,65]

Because of the poor repair potential of articular cartilage, research has focused on methods of facilitating repair of hyaline cartilage. Surgical debridement continues to play an important role in treatment of articular surface defects. Arthroscopic shaving of partial-thickness defects essentially relieves clinical signs associated with defects; however, it does not appear to dramatically alter healing potential.[62,63] Recent advances have led to the concept of perforation or abrasion of the calcified cartilage layer and invasion to the subchondral region to elicit vascular invasion and a local repair response.[63,66] Another important approach to repair of articular defects is the concept of early continuous passive motion postoperatively, because prolonged immobilization impairs the biomechanical and biochemical healing capacity of articular cartilage.[62,63,67] Additional research includes the potential for electrical stimulation, cartilage grafting, and the effects of growth factors on cartilage healing.[63]

Advances for the Future

Wound healing remains an area with tremendous research potential. An exciting area of interest includes the potential for modulation of wound healing with the use of various growth factors. Growth factors are low-molecular-weight polypeptide molecules that are located throughout the body and have a powerful influence on cellular activity. Factors such as transforming growth factor alpha and beta, platelet-derived growth factor, fibroblast growth factor, epidermal growth factor, and insulin-like growth factor modulate cellular activity by binding to receptors on appropriate target cells, resulting in stimulation of specific cellular activities. These substances may have various effects, including chemotaxis, mitogenesis, cellular activation, and stimulation of all aspects of wound healing, with their exogenous use intended to expedite the repair of injured tissues.[68-70] Recently, the potential for stem cell application to healing tissues has generated a great deal of interest. In addition to the altered cellular environment generated by the cells, the potential for stem cells to grow and differentiate into cells specific to a particular tissue type is exciting.

Summary

Wound healing in most tissues involves an initial process of hemostasis followed by inflammation. This allows for debridement of the injured region followed by chemotaxis of the appropriate cells for production of repair tissue. In general, the repair process involves production of collagen, bone, or myofibrils. Maturation occurs over time, allowing for appropriate alignment of collagen and tissue for resistance of tensile forces. The initiation of physical rehabilitation in the postoperative period is important, and the techniques should be used to augment the maturation phase to maximize early return of tensile strength and function. The time of initiation of various therapeutic modalities is important, and knowledge of the basic patterns of wound healing is imperative.

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